

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

Complete

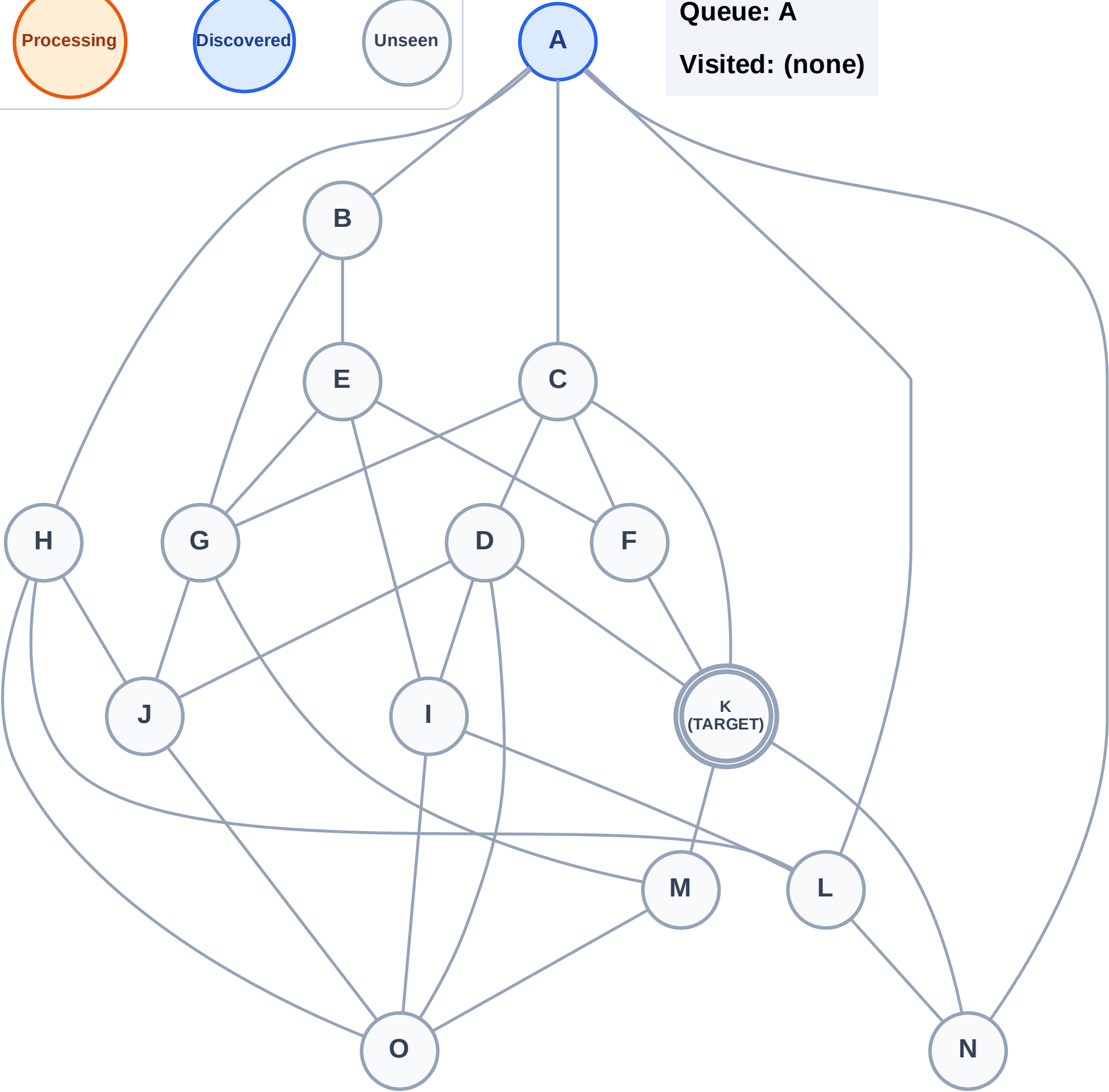
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

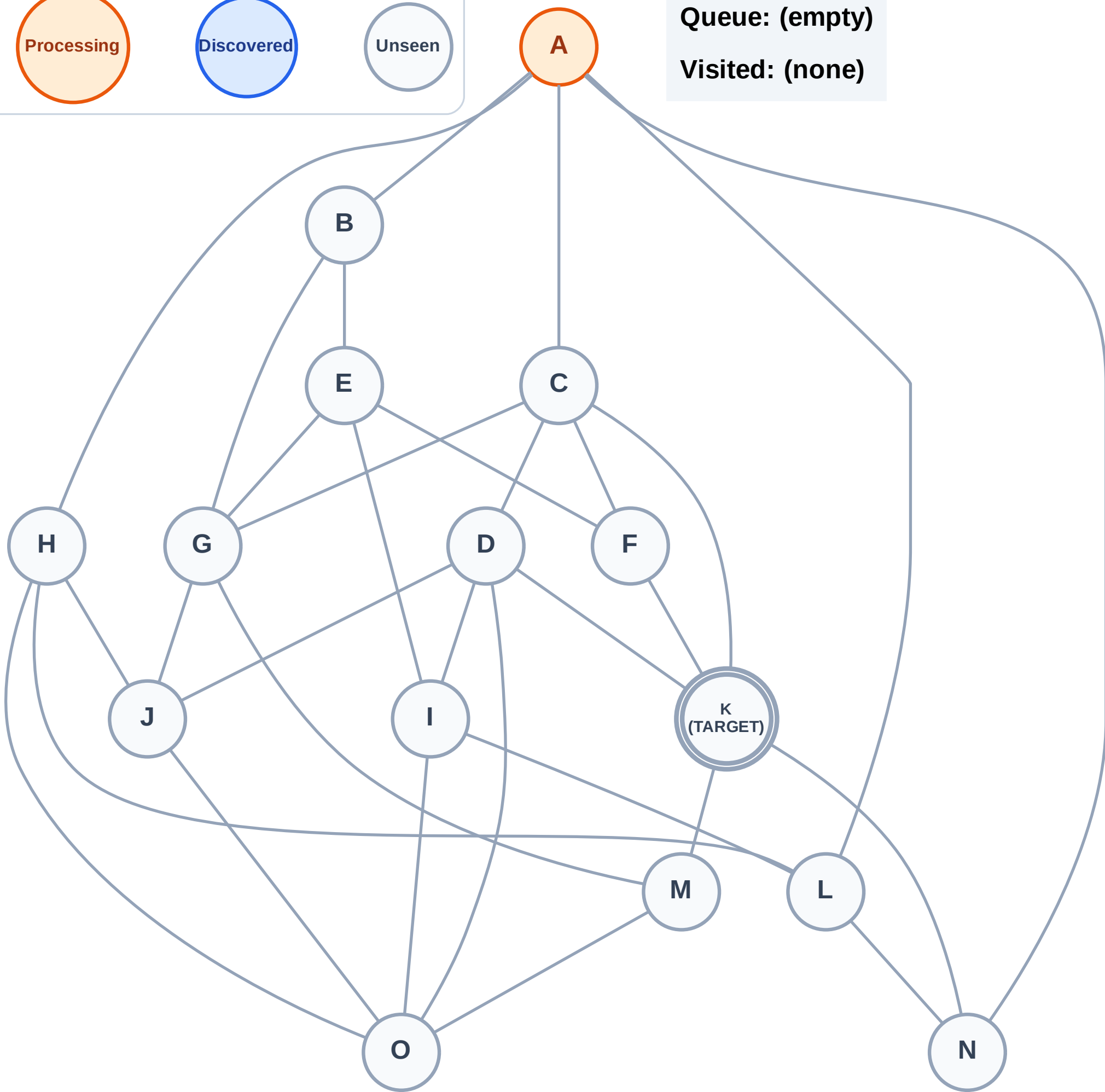
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

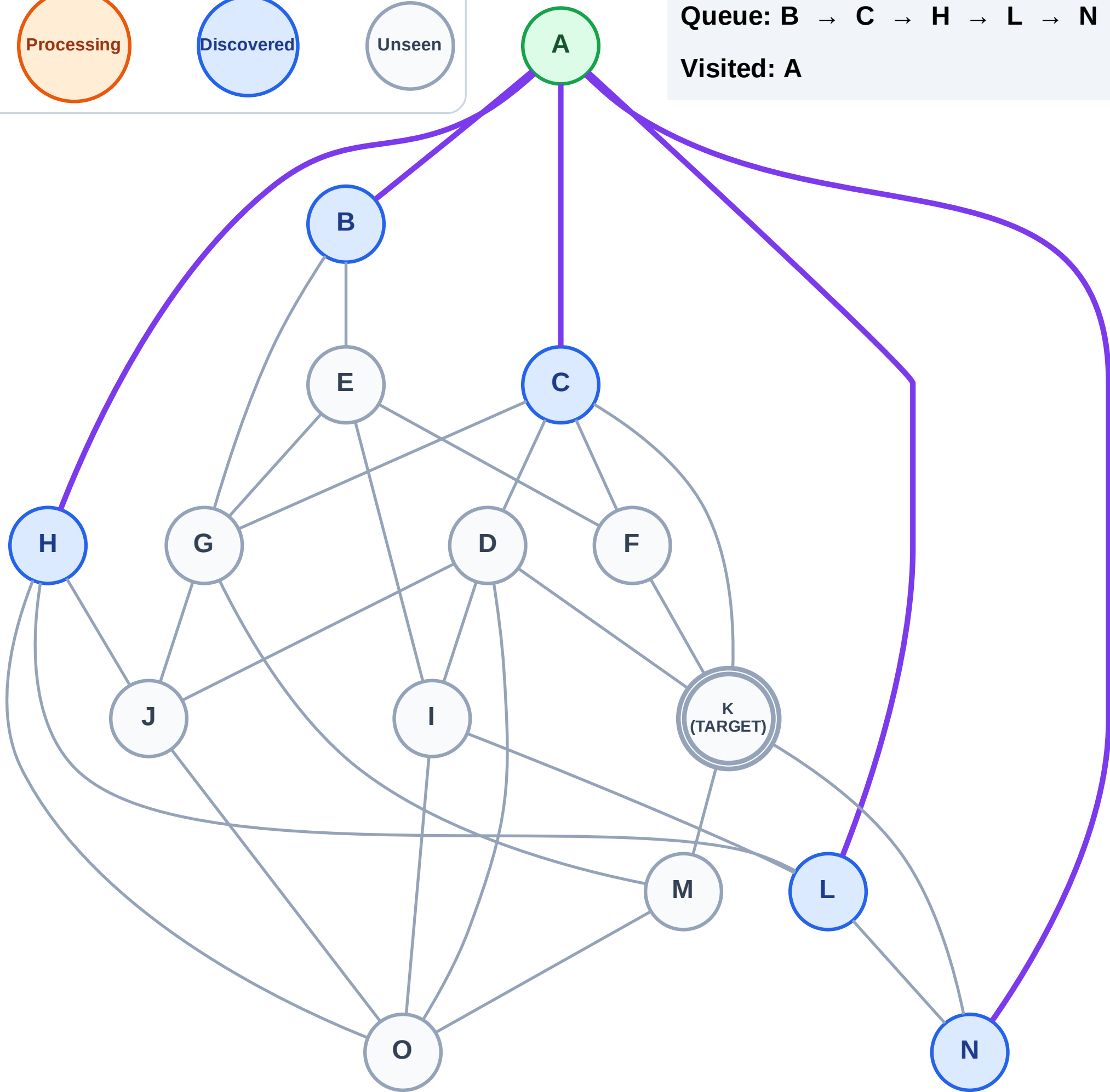
Step 3: Discover B, C, H, L, N from A

Legend



Queue: B → C → H → L → N

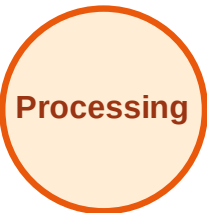
Visited: A



BFS Traversal

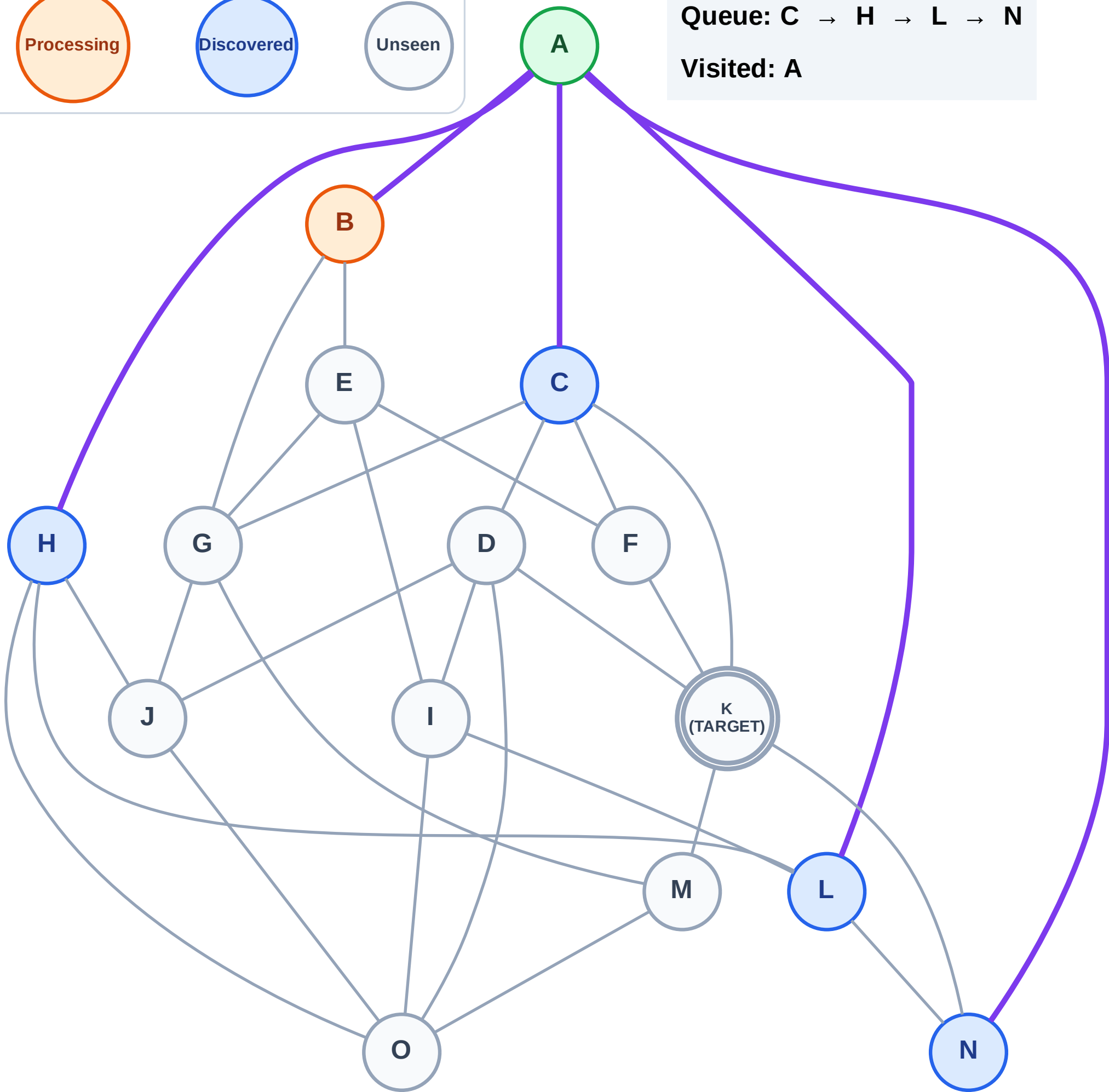
Step 4: Process node B

Legend



Queue: C → H → L → N

Visited: A



BFS Traversal

Step 5: Discover E, G from B

Legend

Complete

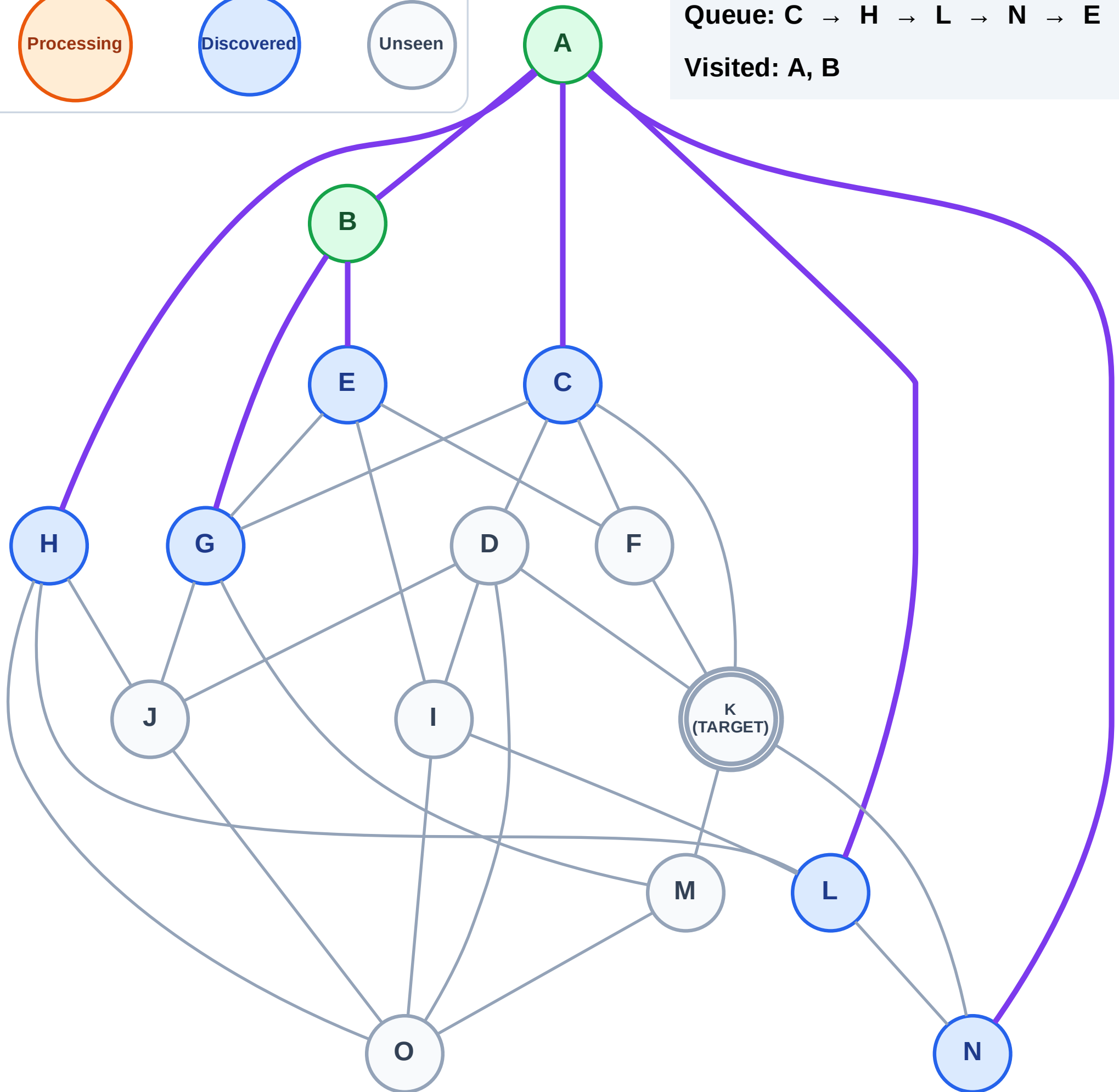
Processing

Discovered

Unseen

Queue: C → H → L → N → E → G

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



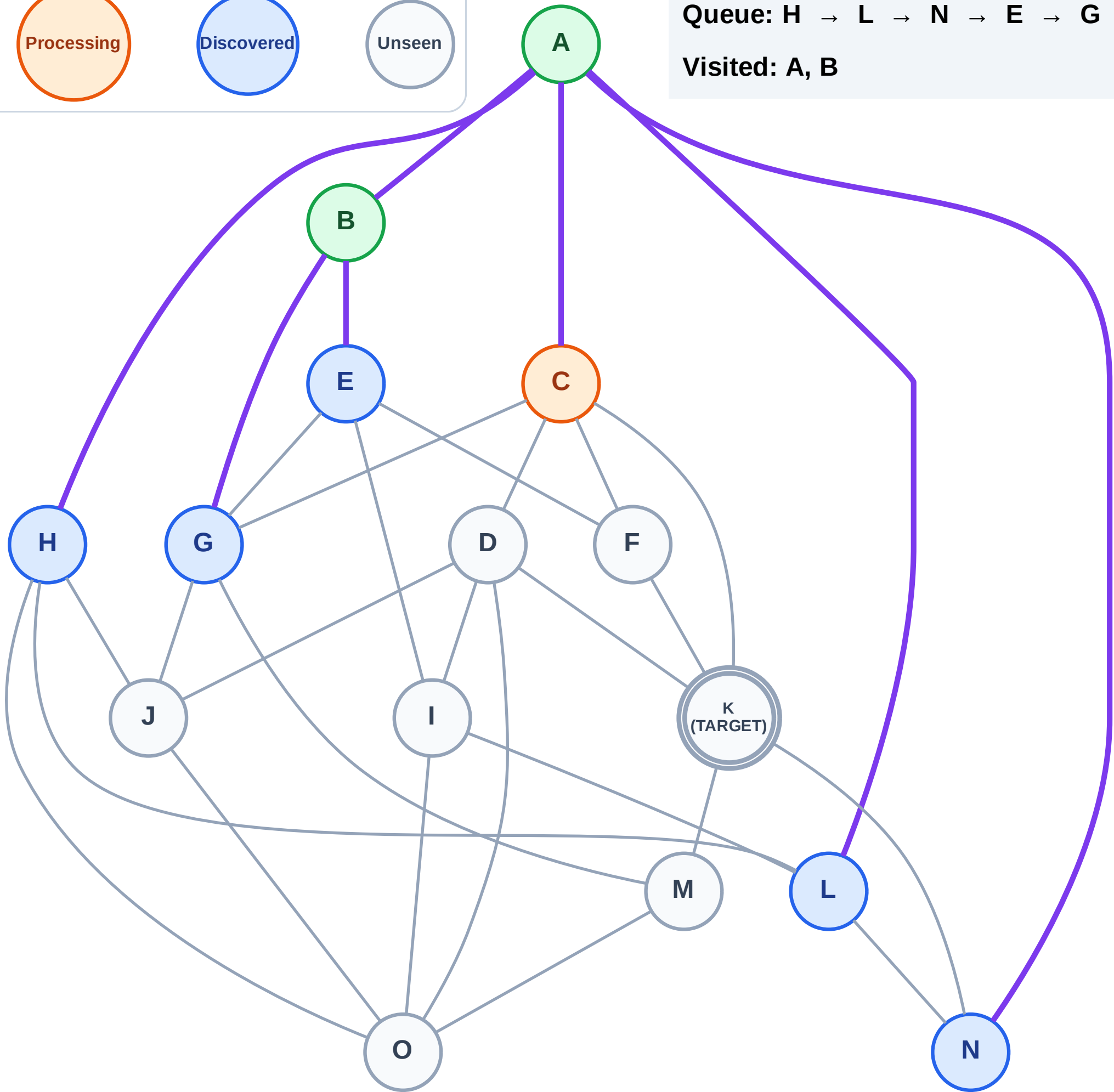
Discovered



Unseen

Queue: H → L → N → E → G

Visited: A, B



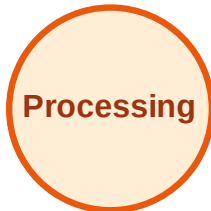
BFS Traversal

Step 7: Discover D, F, K from C

Legend



Complete



Processing



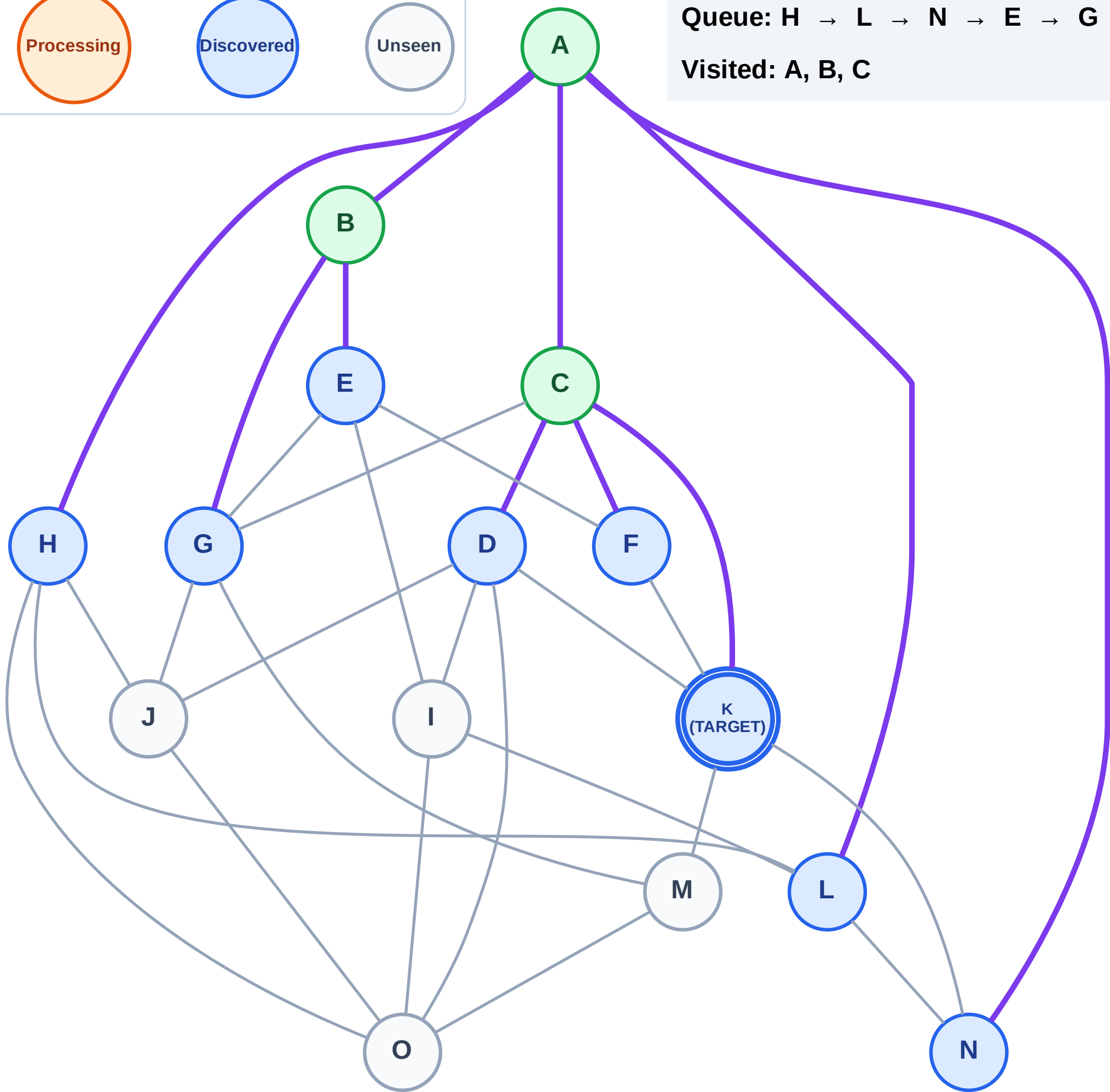
Discovered



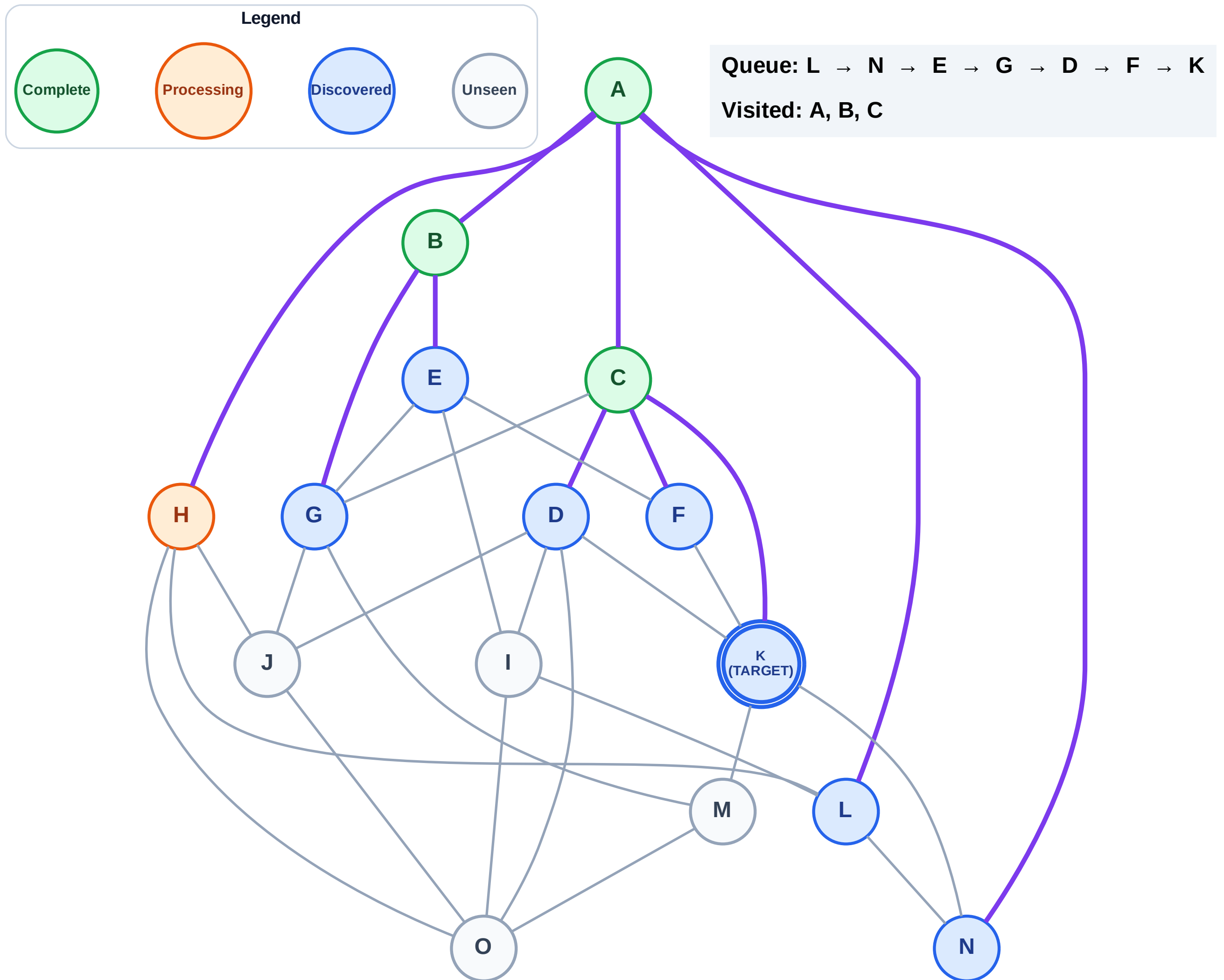
Unseen

Queue: H → L → N → E → G → D → F → K

Visited: A, B, C



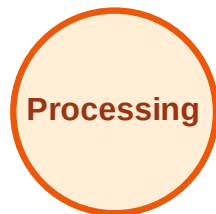
Step 8: Process node H



BFS Traversal

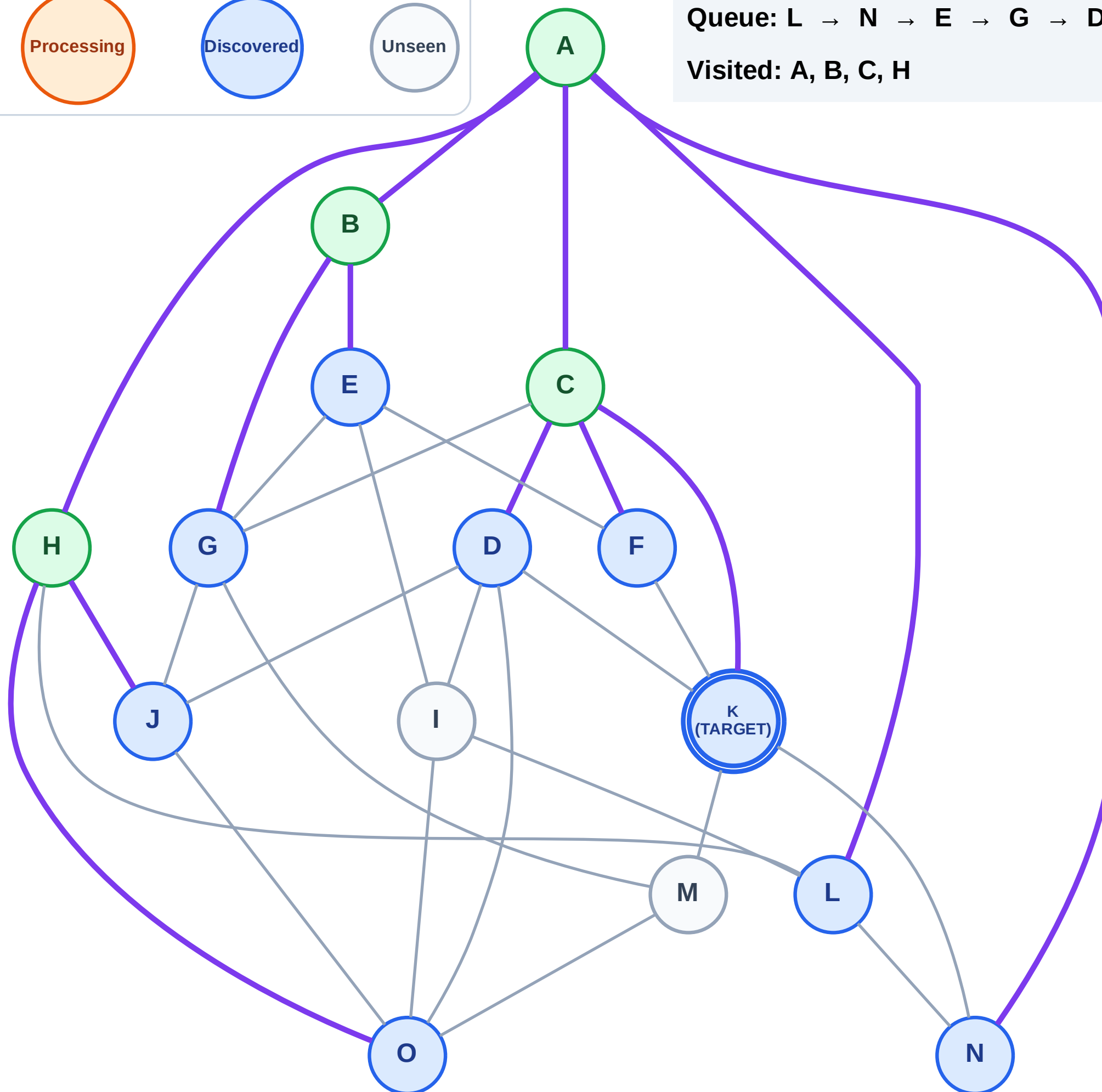
Step 9: Discover J, O from H

Legend



Queue: L → N → E → G → D → F → K → J → O

Visited: A, B, C, H



BFS Traversal

Step 10: Process node L

Legend



Complete



Processing



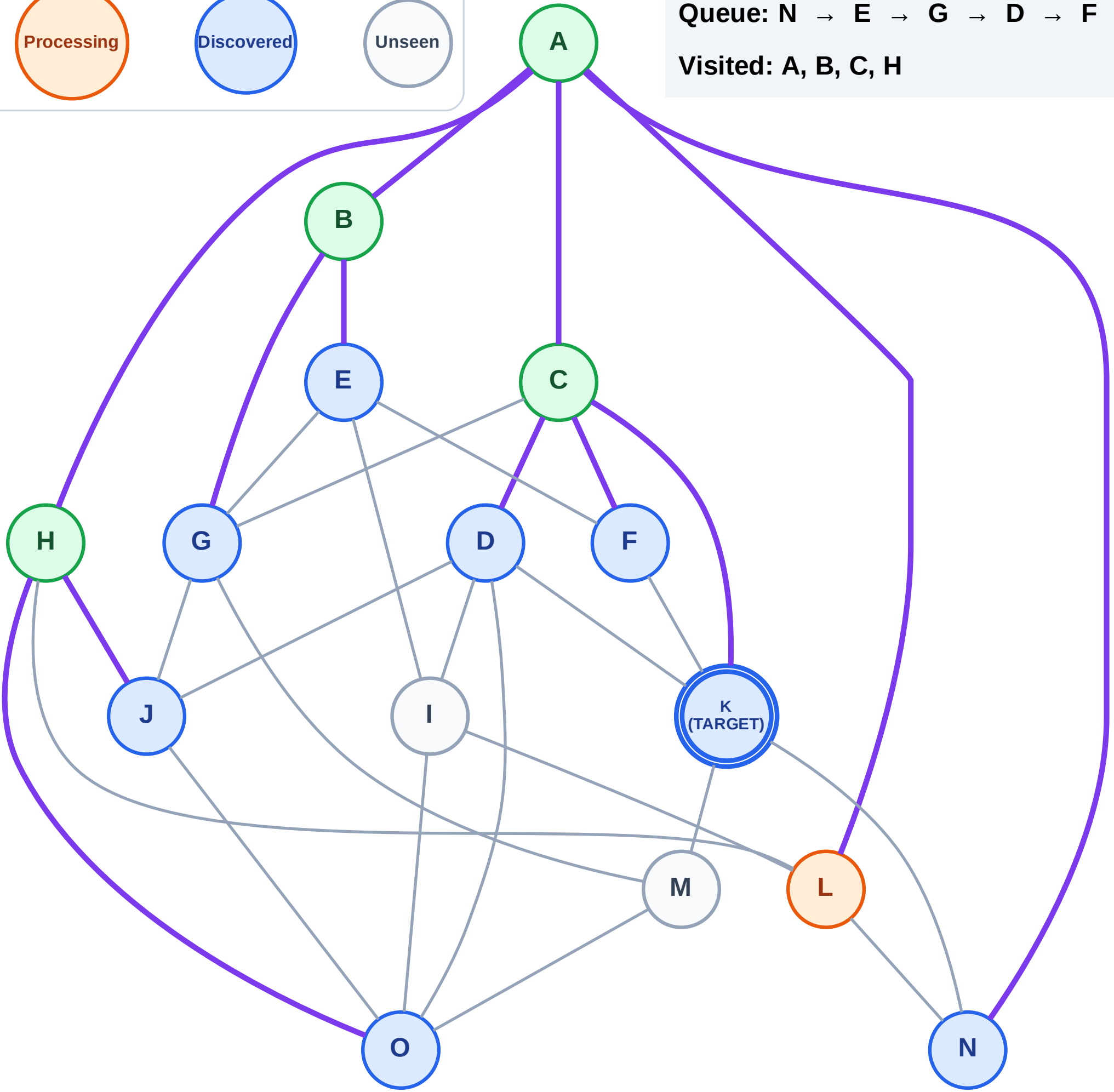
Discovered



Unseen

Queue: N → E → G → D → F → K → J → O

Visited: A, B, C, H



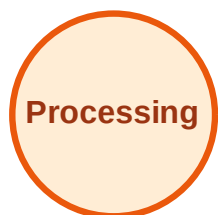
BFS Traversal

Step 11: Discover I from L

Legend



Complete



Processing



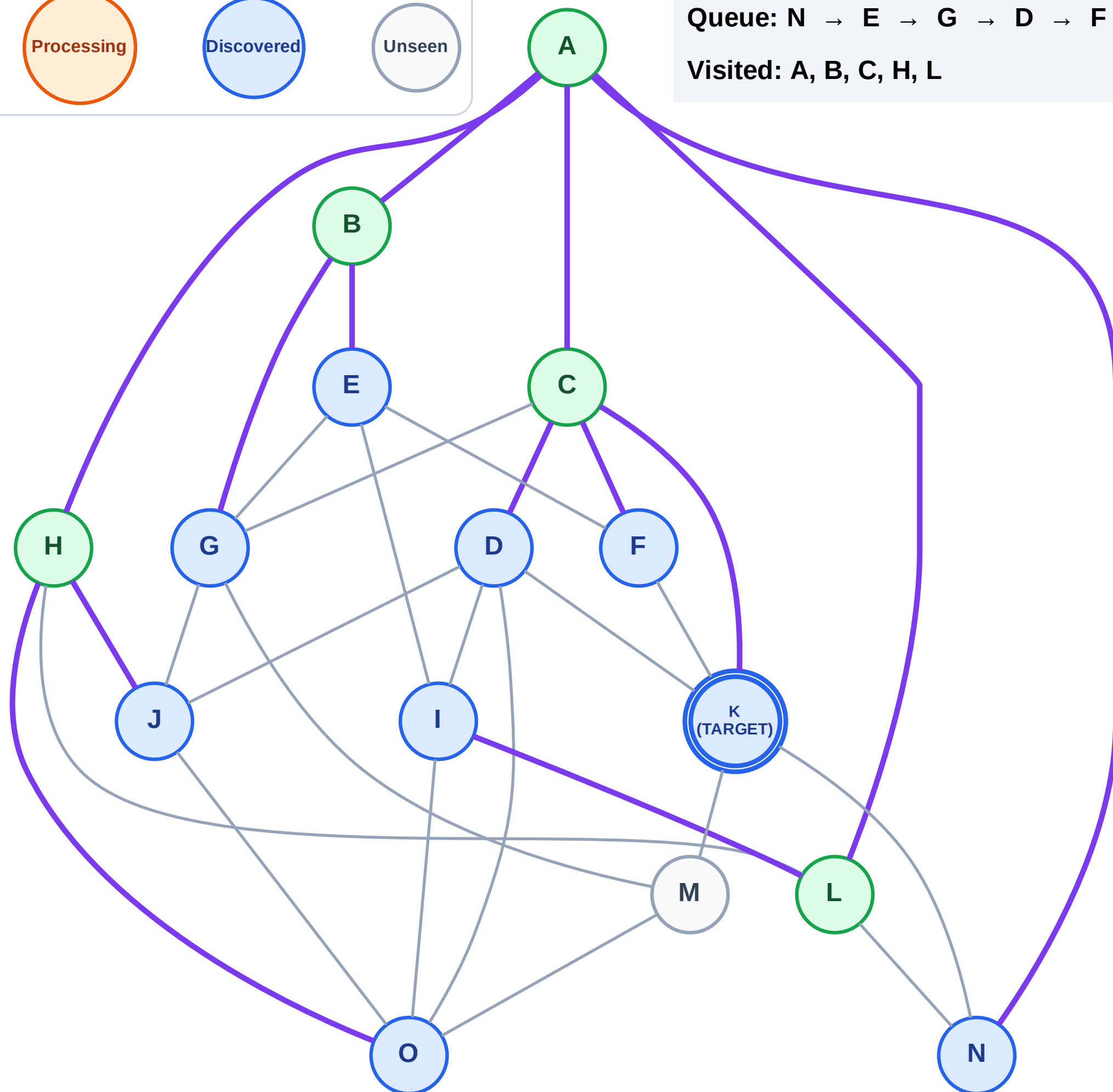
Discovered



Unseen

Queue: N → E → G → D → F → K → J → O → I

Visited: A, B, C, H, L



BFS Traversal

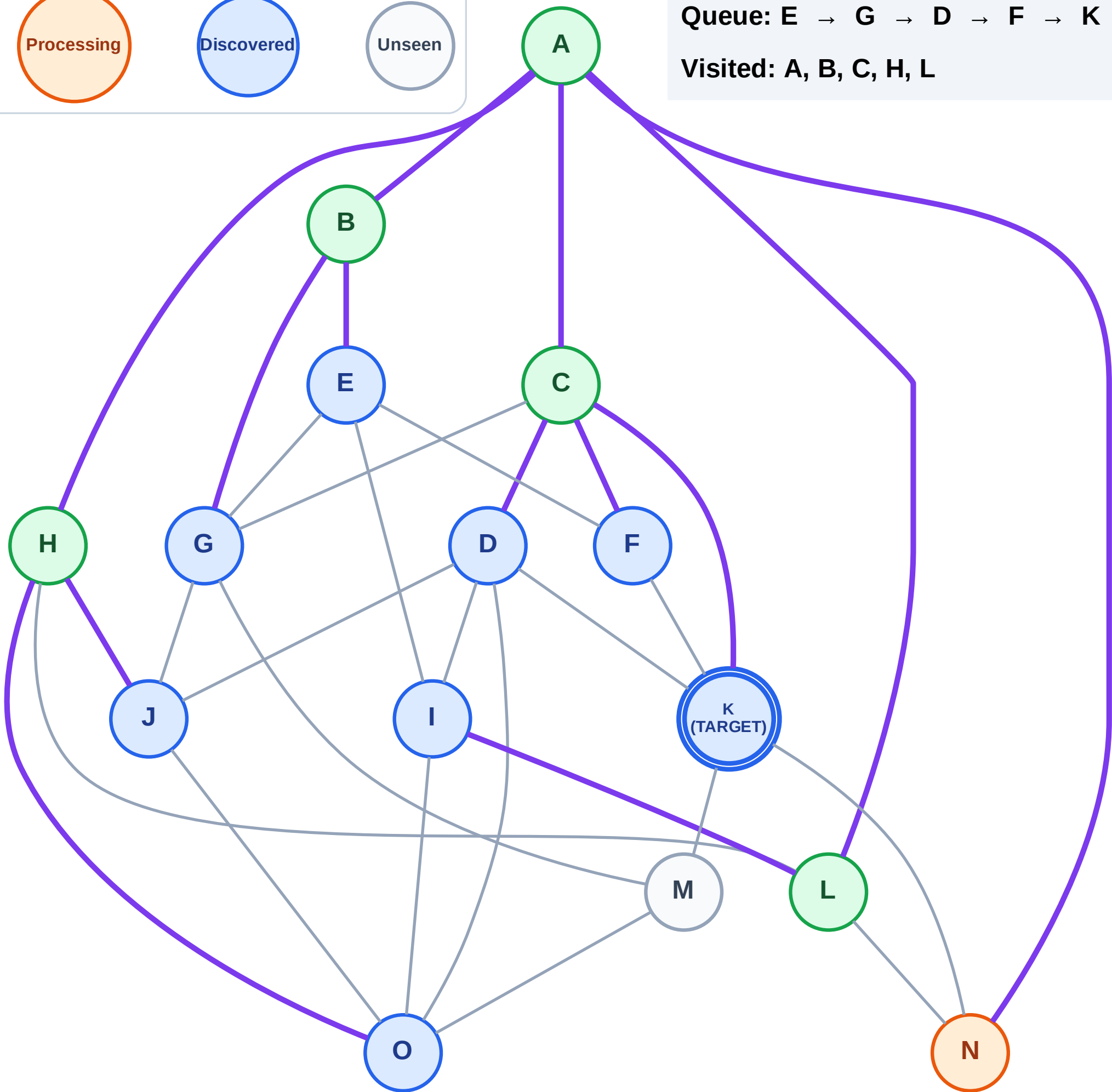
Step 12: Process node N

Legend



Queue: E → G → D → F → K → J → O → I

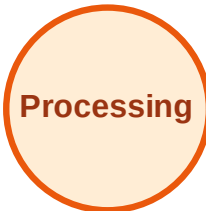
Visited: A, B, C, H, L



BFS Traversal

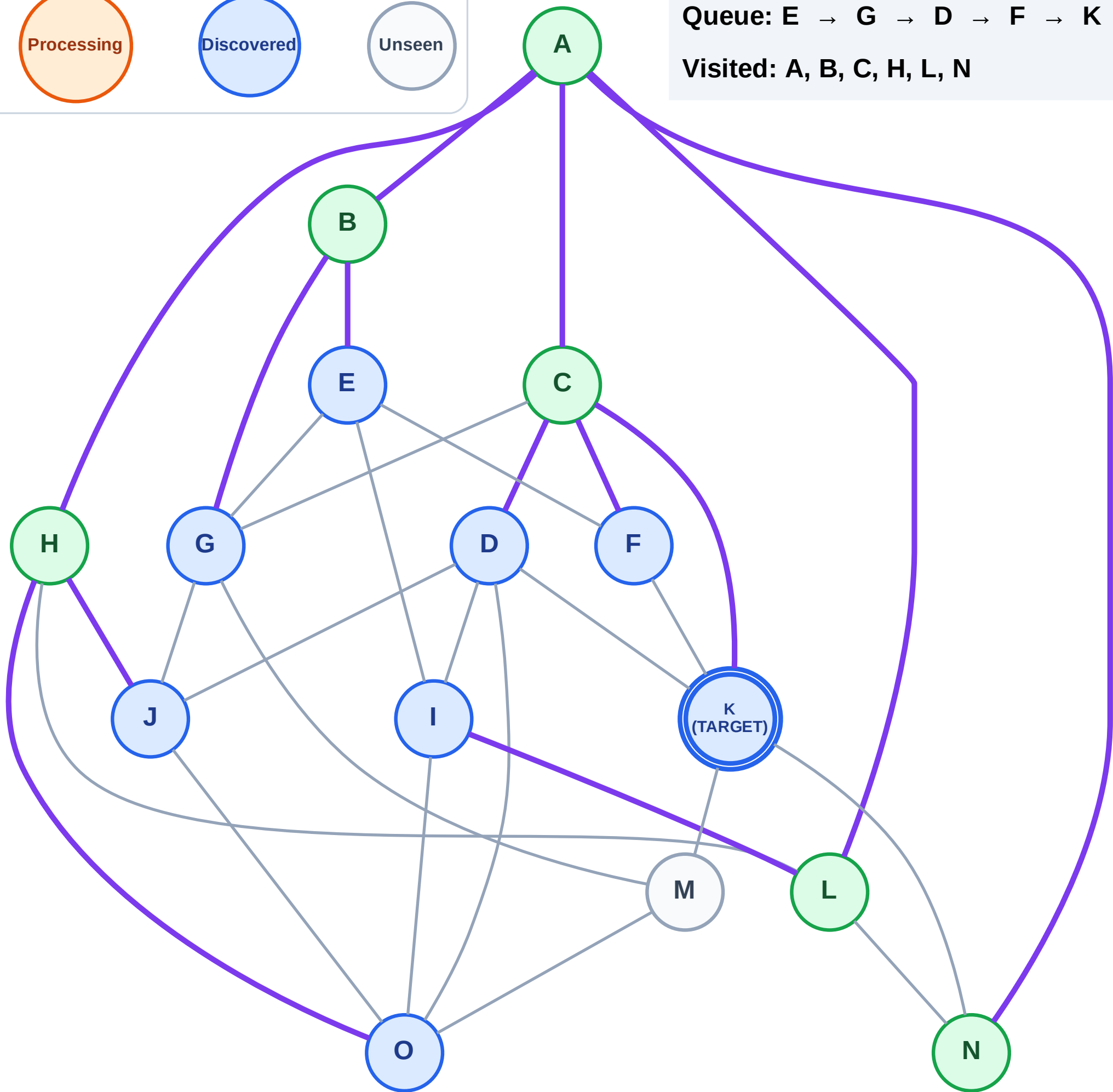
Step 13: No new neighbors from N

Legend



Queue: E → G → D → F → K → J → O → I

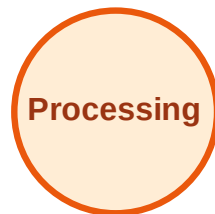
Visited: A, B, C, H, L, N



BFS Traversal

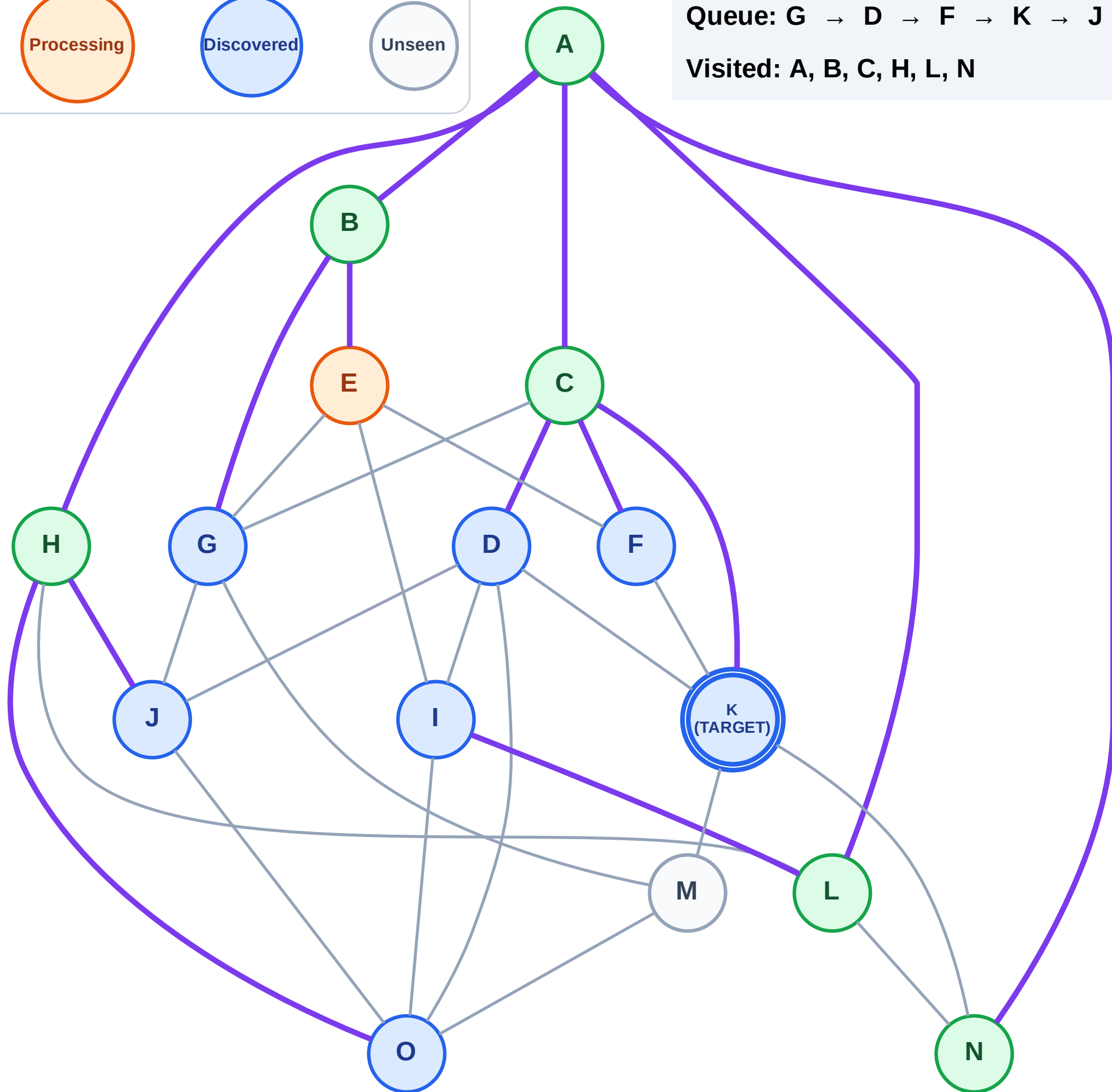
Step 14: Process node E

Legend



Queue: G → D → F → K → J → O → I

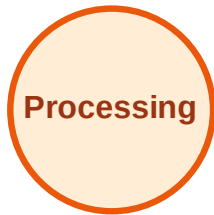
Visited: A, B, C, H, L, N



BFS Traversal

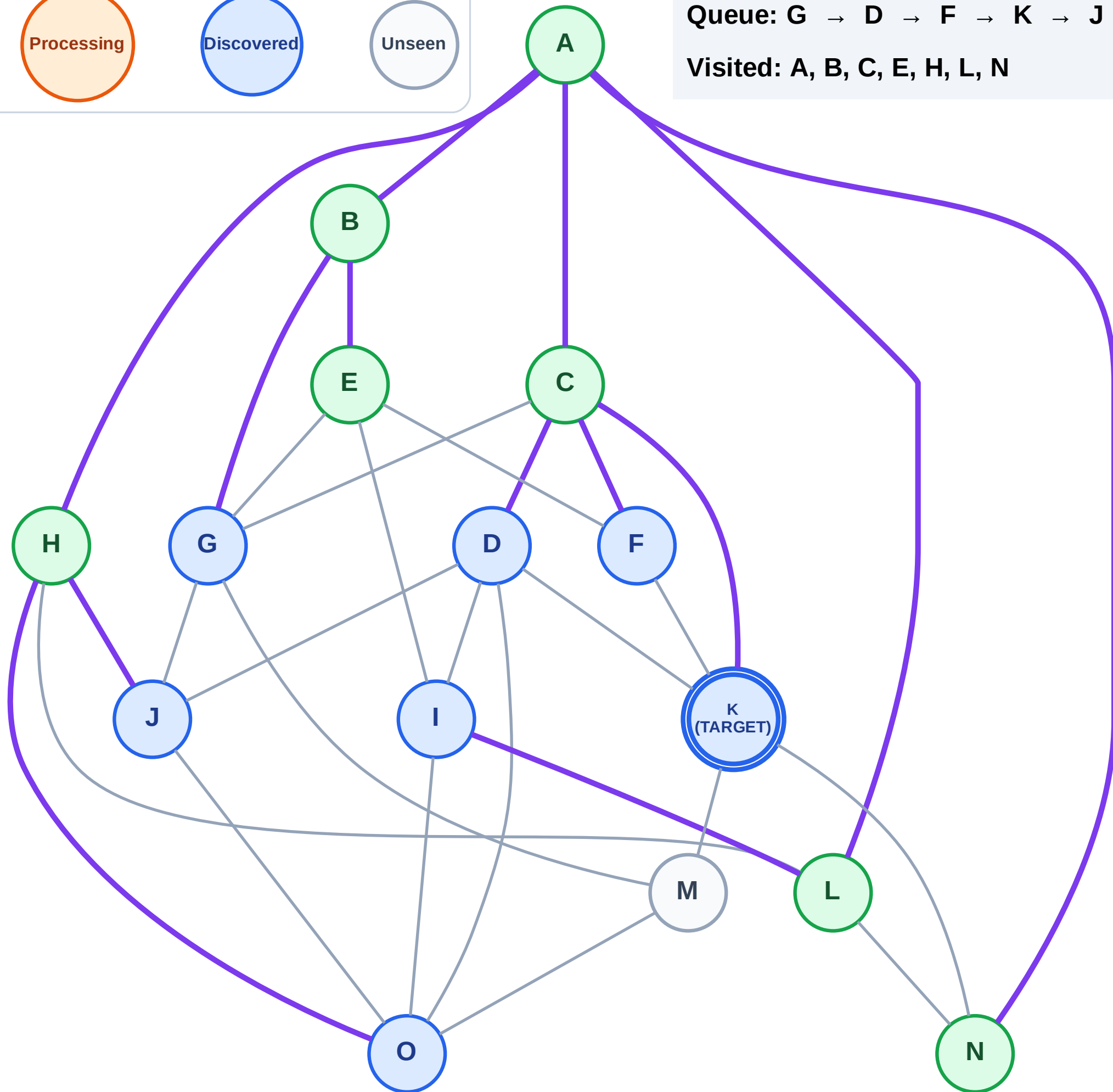
Step 15: No new neighbors from E

Legend



Queue: G → D → F → K → J → O → I

Visited: A, B, C, E, H, L, N



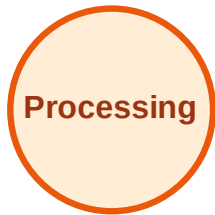
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



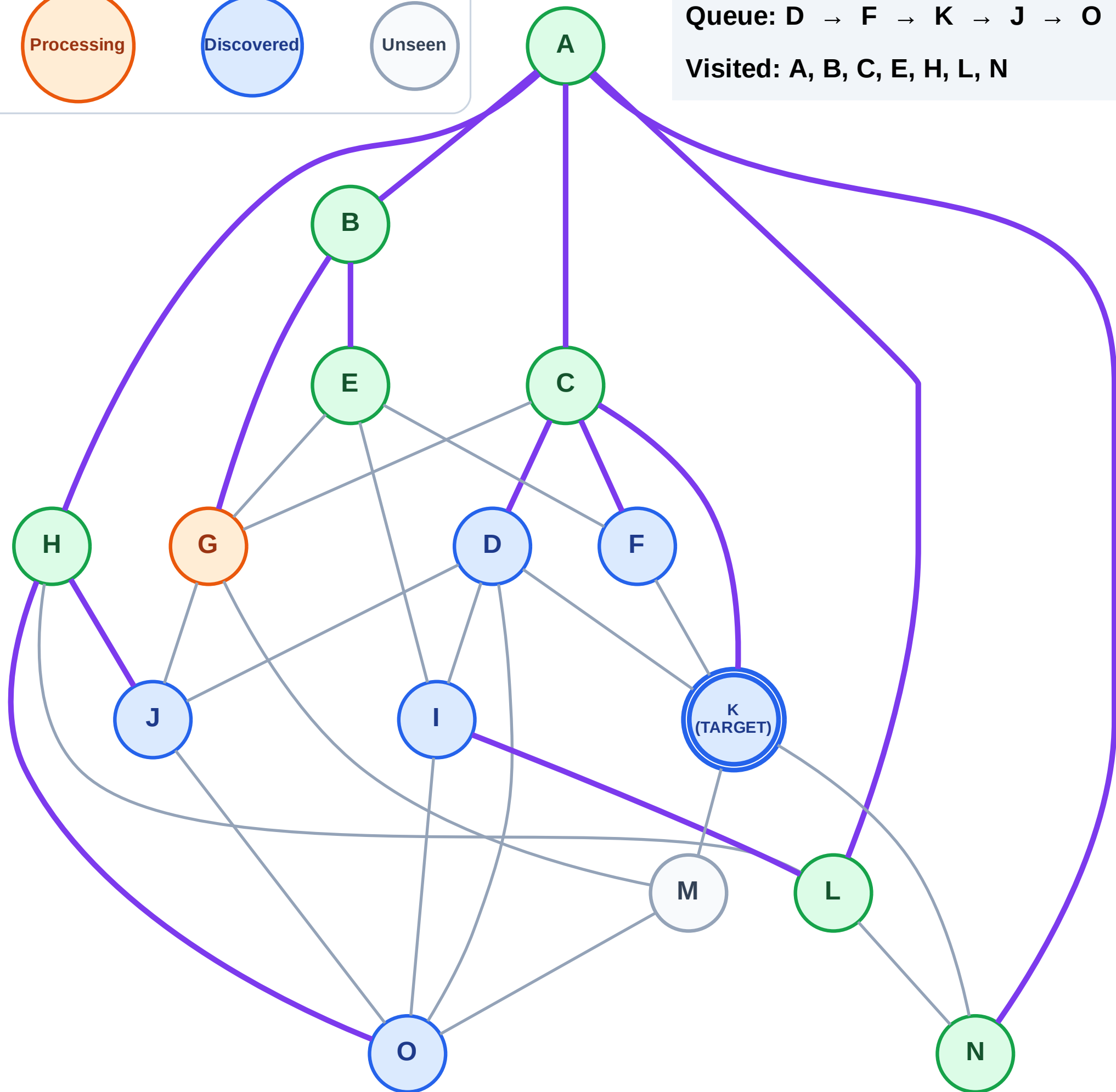
Discovered



Unseen

Queue: D → F → K → J → O → I

Visited: A, B, C, E, H, L, N



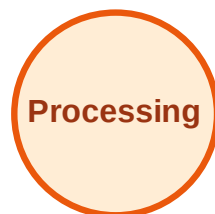
BFS Traversal

Step 17: Discover M from G

Legend



Complete



Processing



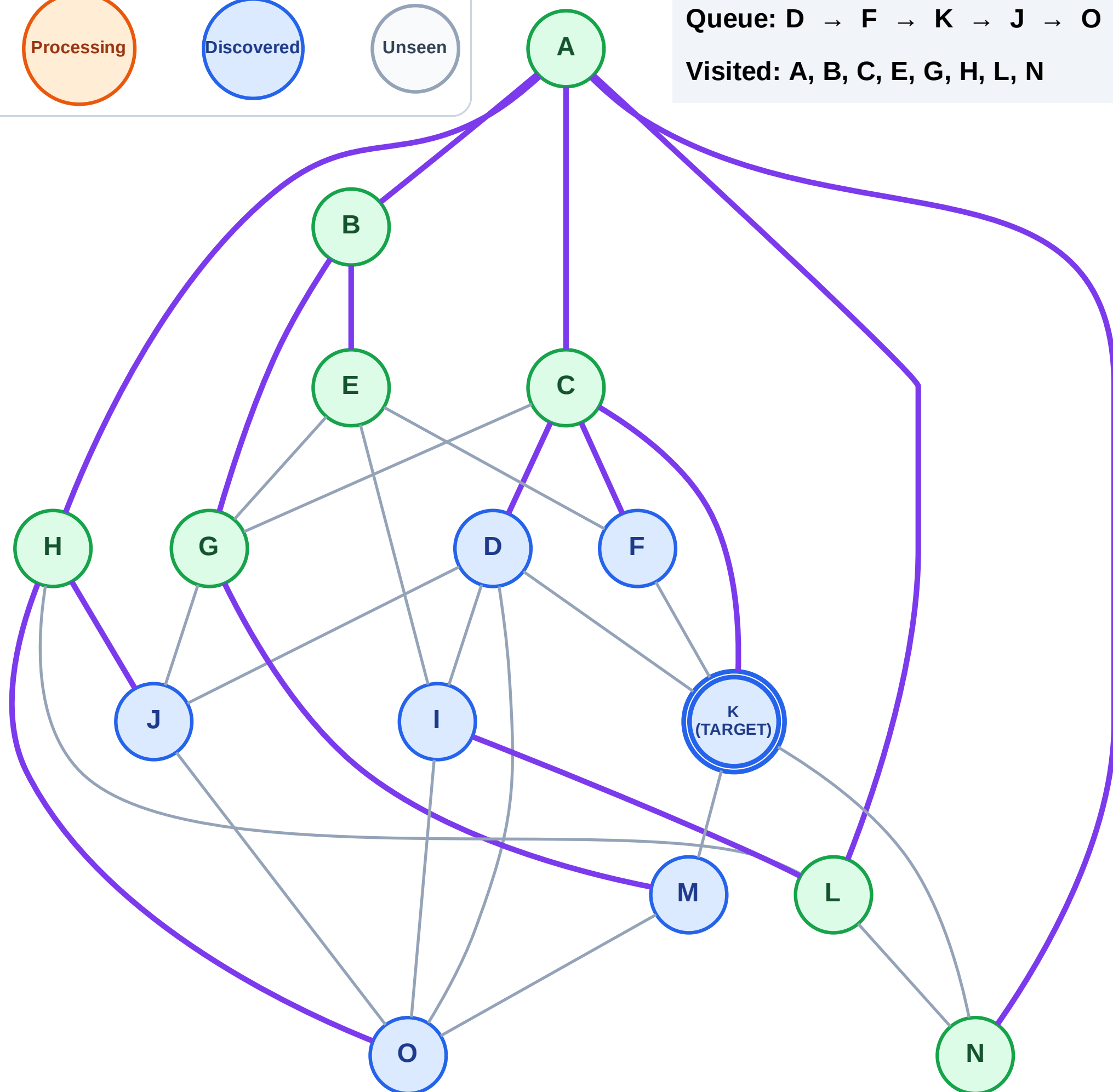
Discovered



Unseen

Queue: D → F → K → J → O → I → M

Visited: A, B, C, E, G, H, L, N



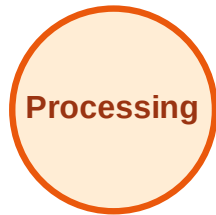
BFS Traversal

Step 18: Process node D

Legend



Complete



Processing



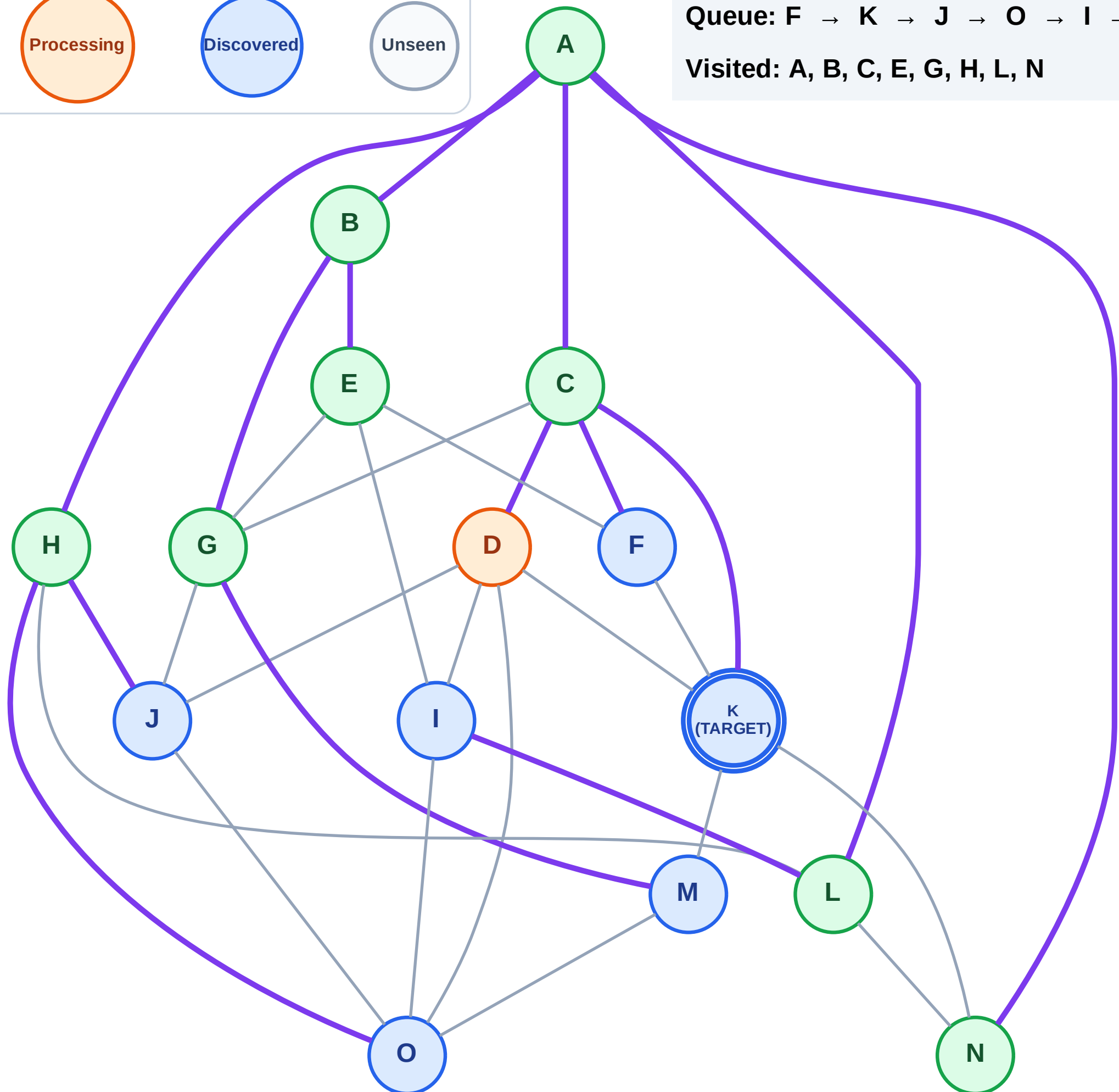
Discovered



Unseen

Queue: F → K → J → O → I → M

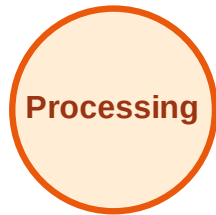
Visited: A, B, C, E, G, H, L, N



BFS Traversal

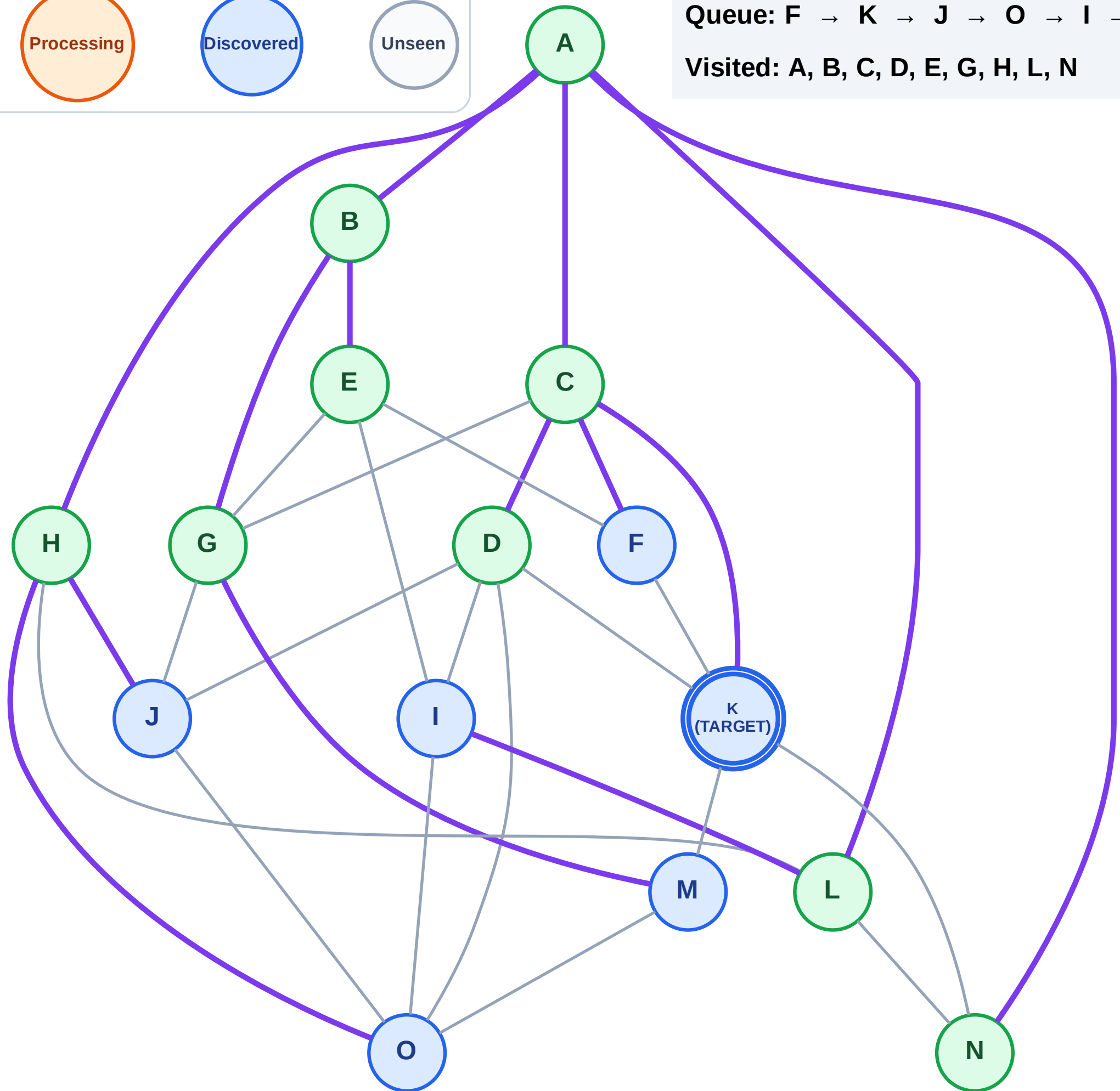
Step 19: No new neighbors from D

Legend



Queue: F → K → J → O → I → M

Visited: A, B, C, D, E, G, H, L, N



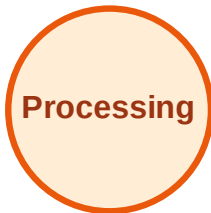
BFS Traversal

Step 20: Process node F

Legend



Complete



Processing



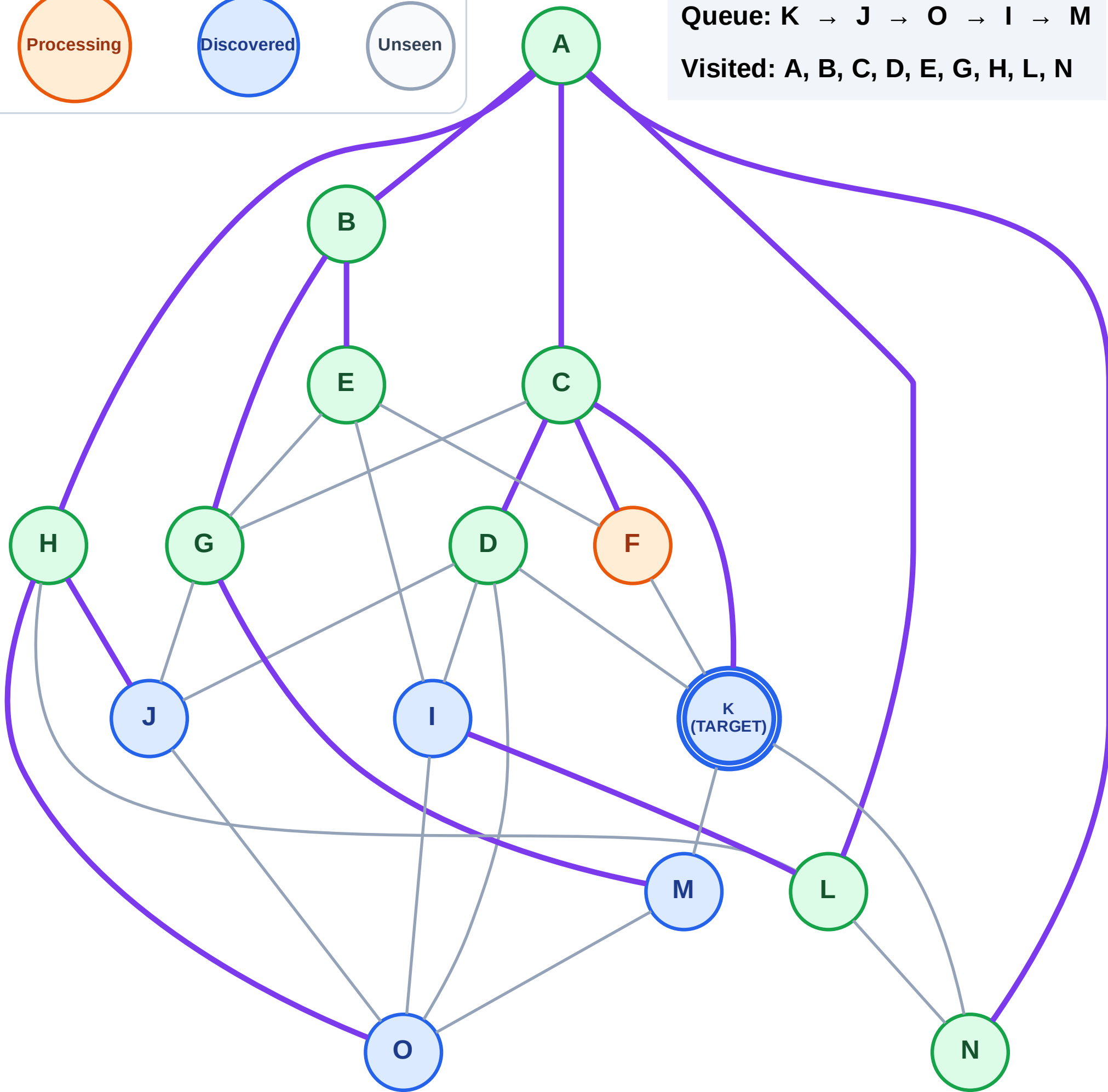
Discovered



Unseen

Queue: K → J → O → I → M

Visited: A, B, C, D, E, G, H, L, N



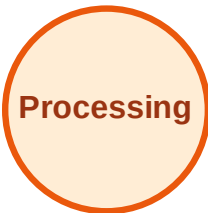
BFS Traversal

Step 21: No new neighbors from F

Legend



Complete



Processing



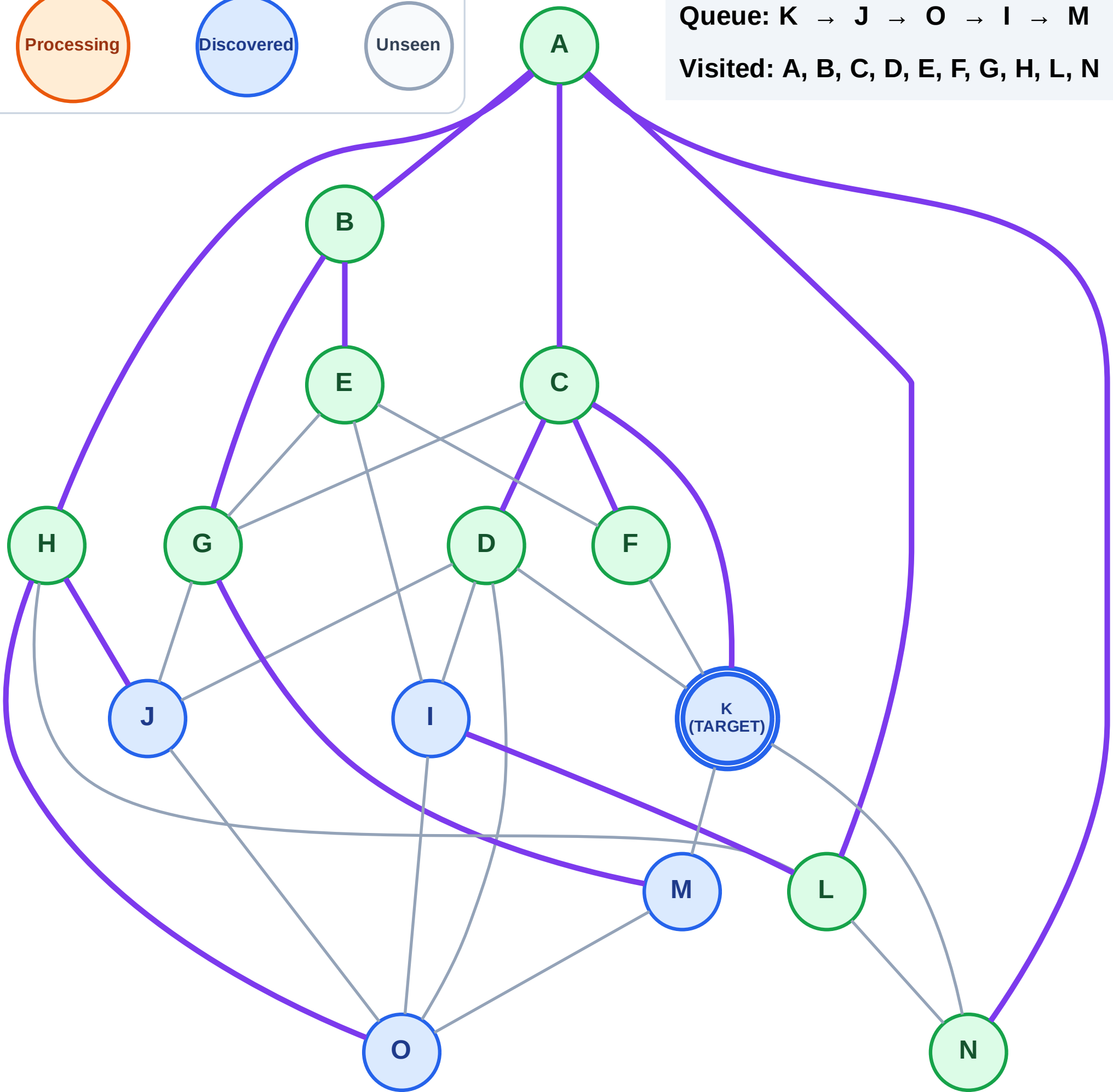
Discovered



Unseen

Queue: K → J → O → I → M

Visited: A, B, C, D, E, F, G, H, L, N



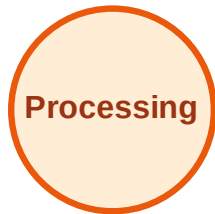
BFS Traversal

Step 22: Process node K

Legend



Complete



Processing



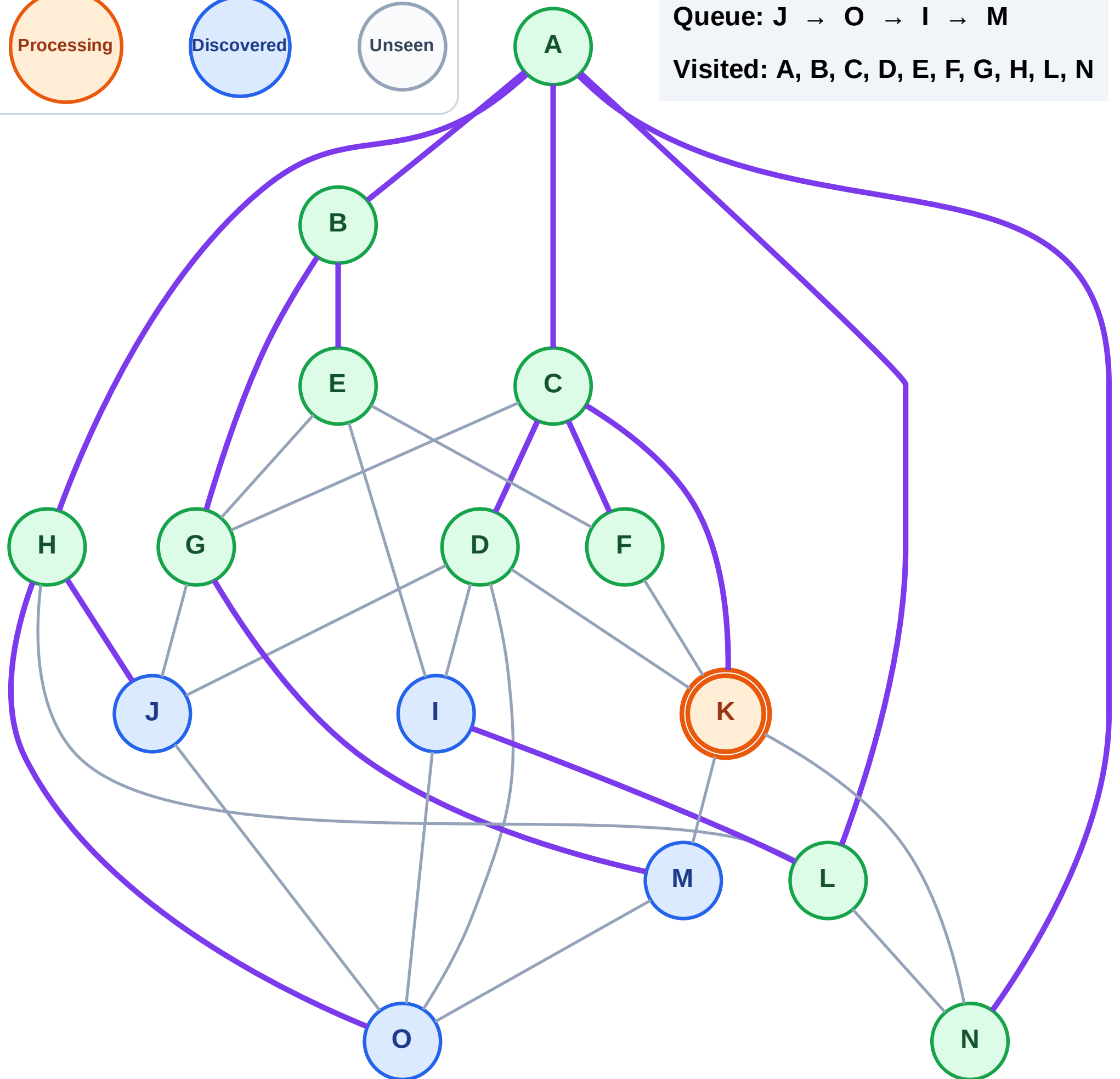
Discovered



Unseen

Queue: J → O → I → M

Visited: A, B, C, D, E, F, G, H, L, N



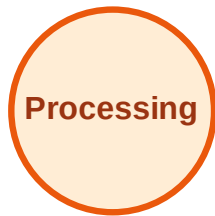
BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → C → K

Legend



Complete



Processing



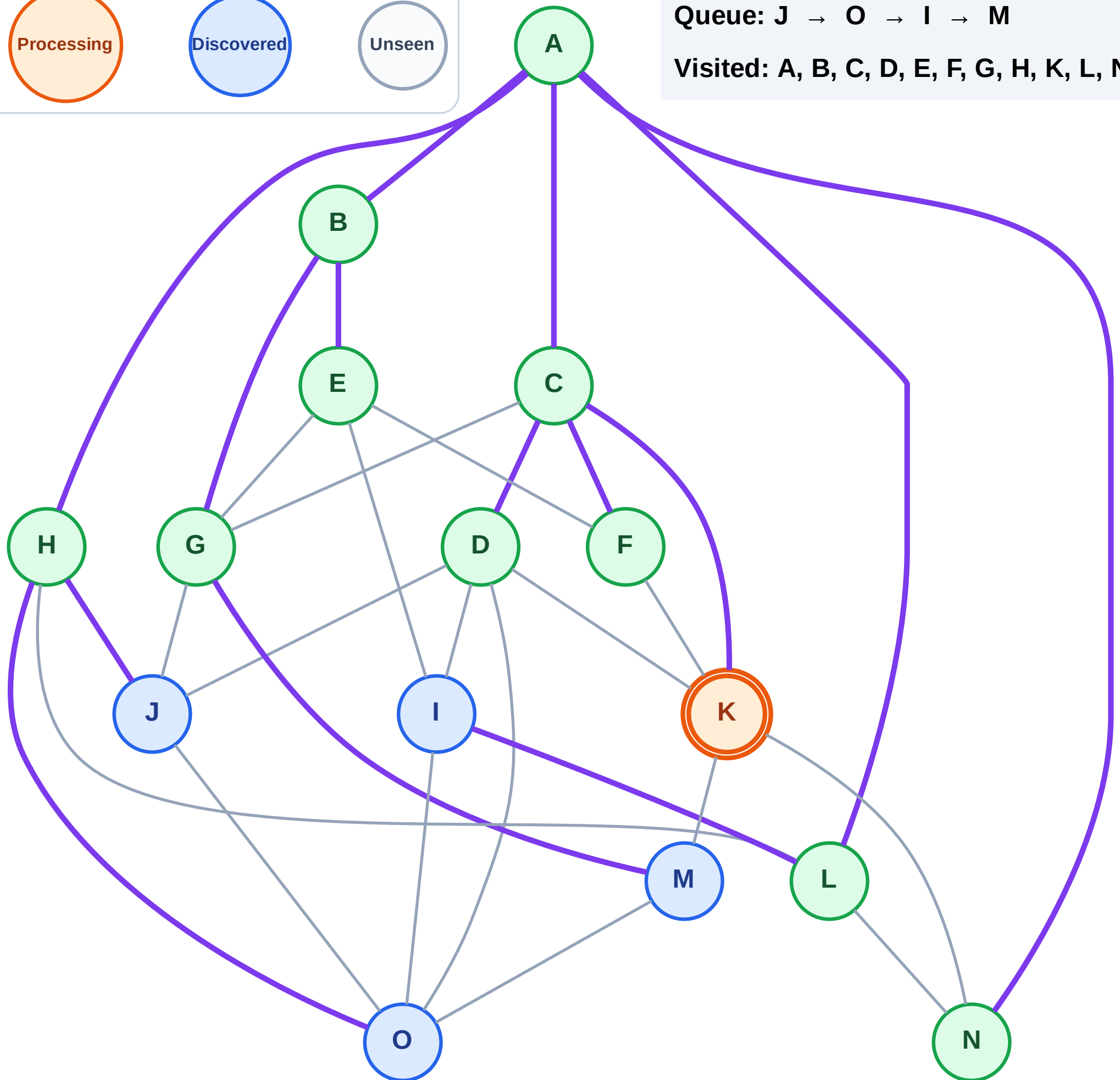
Discovered



Unseen

Queue: J → O → I → M

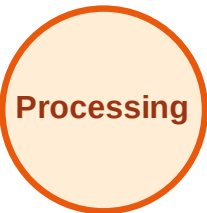
Visited: A, B, C, D, E, F, G, H, K, L, N



DFS Traversal

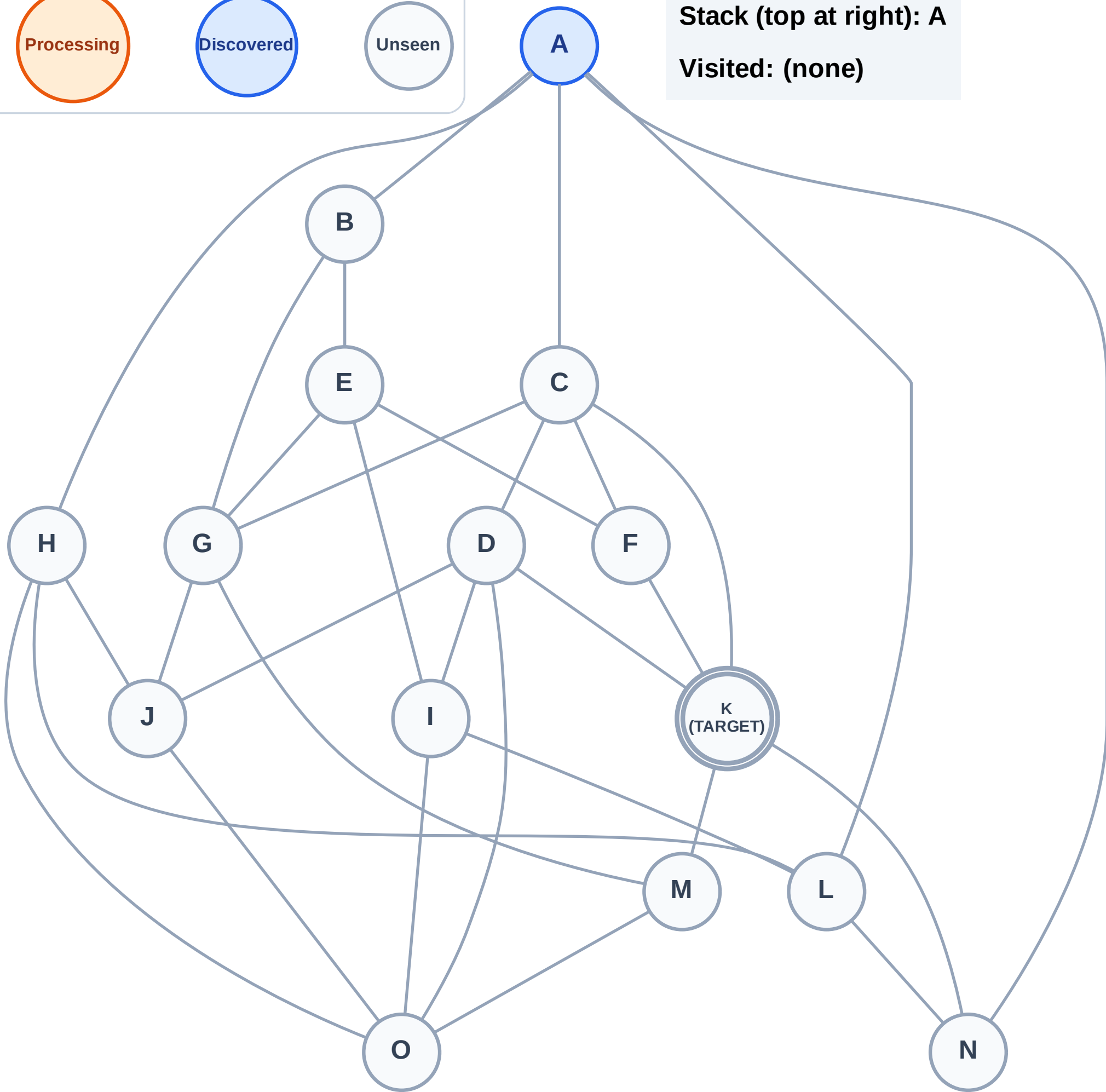
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

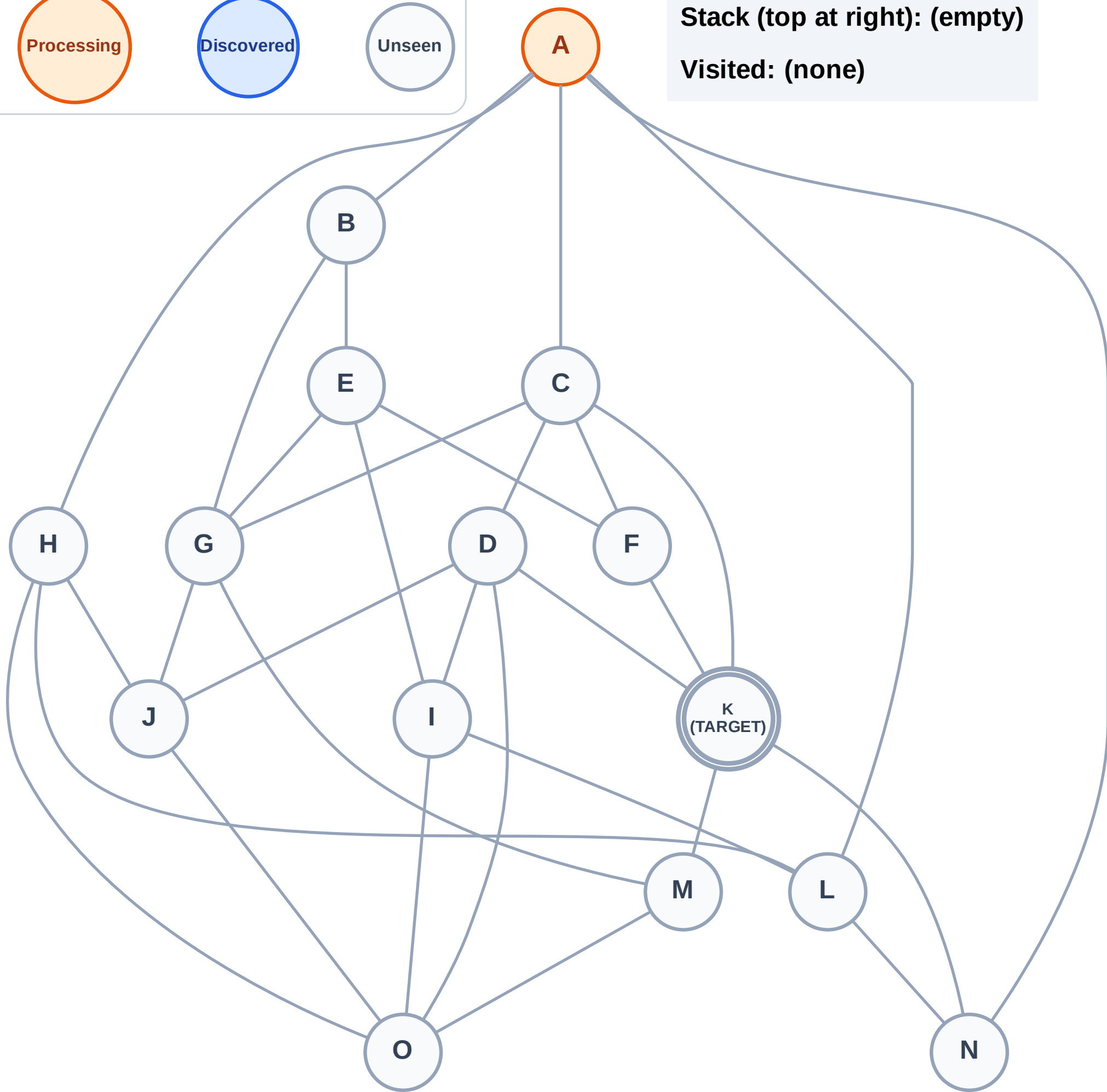
Step 2: Process node A

Legend



Stack (top at right): (empty)

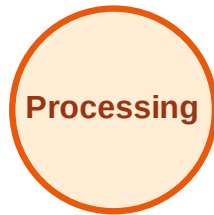
Visited: (none)



DFS Traversal

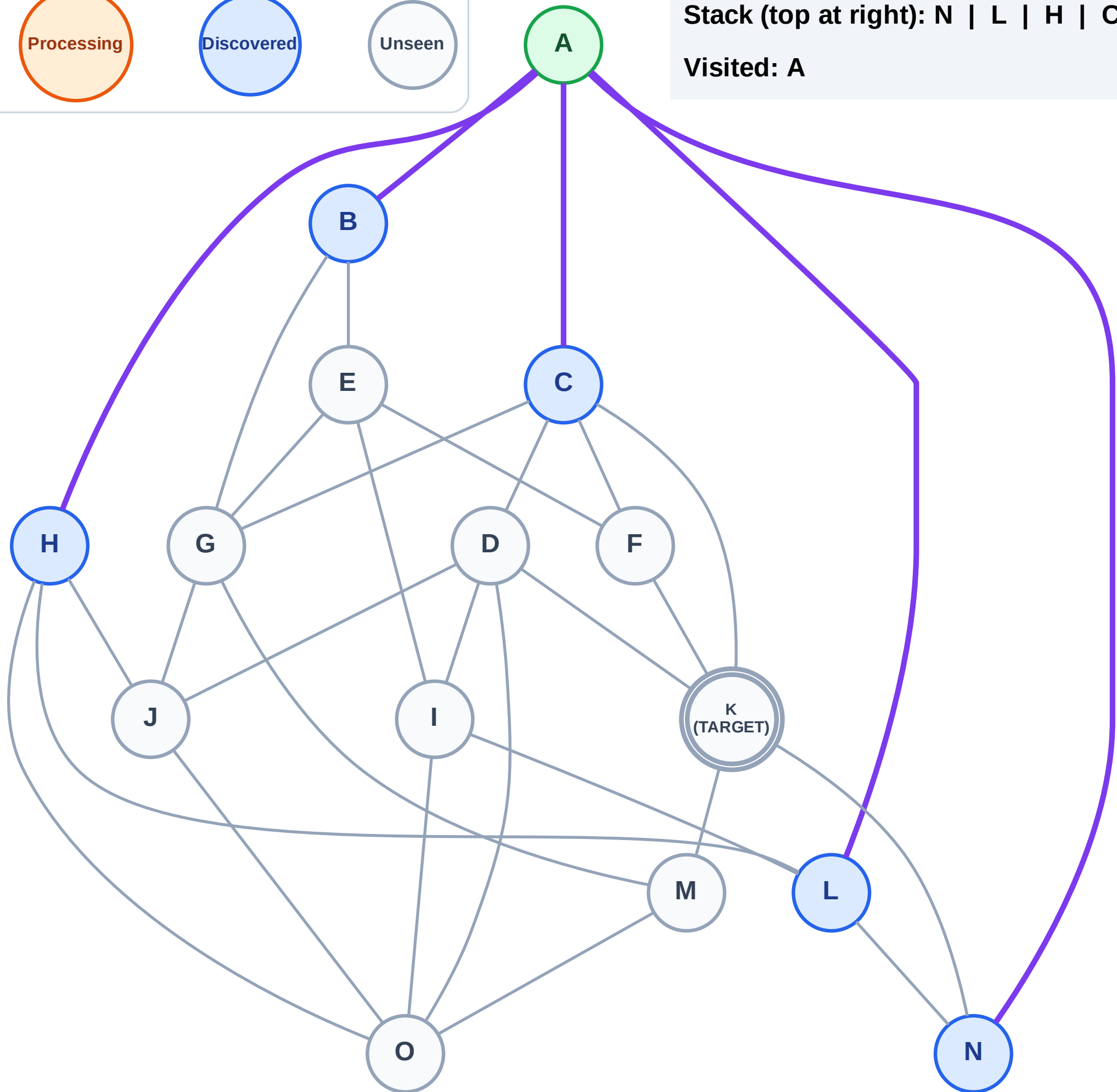
Step 3: Discover N, L, H, C, B from A

Legend



Stack (top at right): N | L | H | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

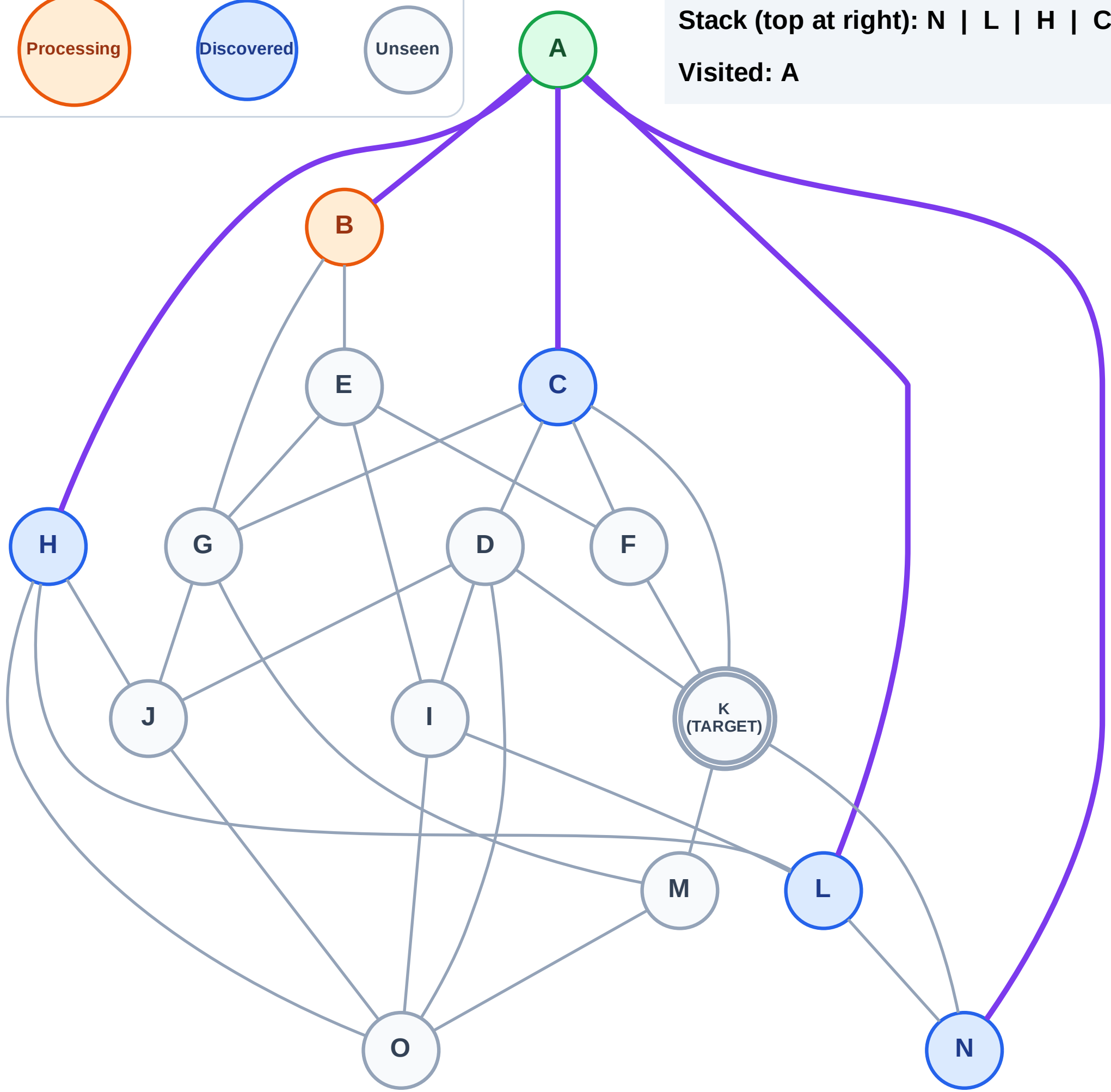
Complete

Processing

Discovered

Unseen

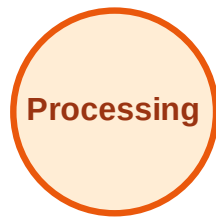
Stack (top at right): N | L | H | C
Visited: A



DFS Traversal

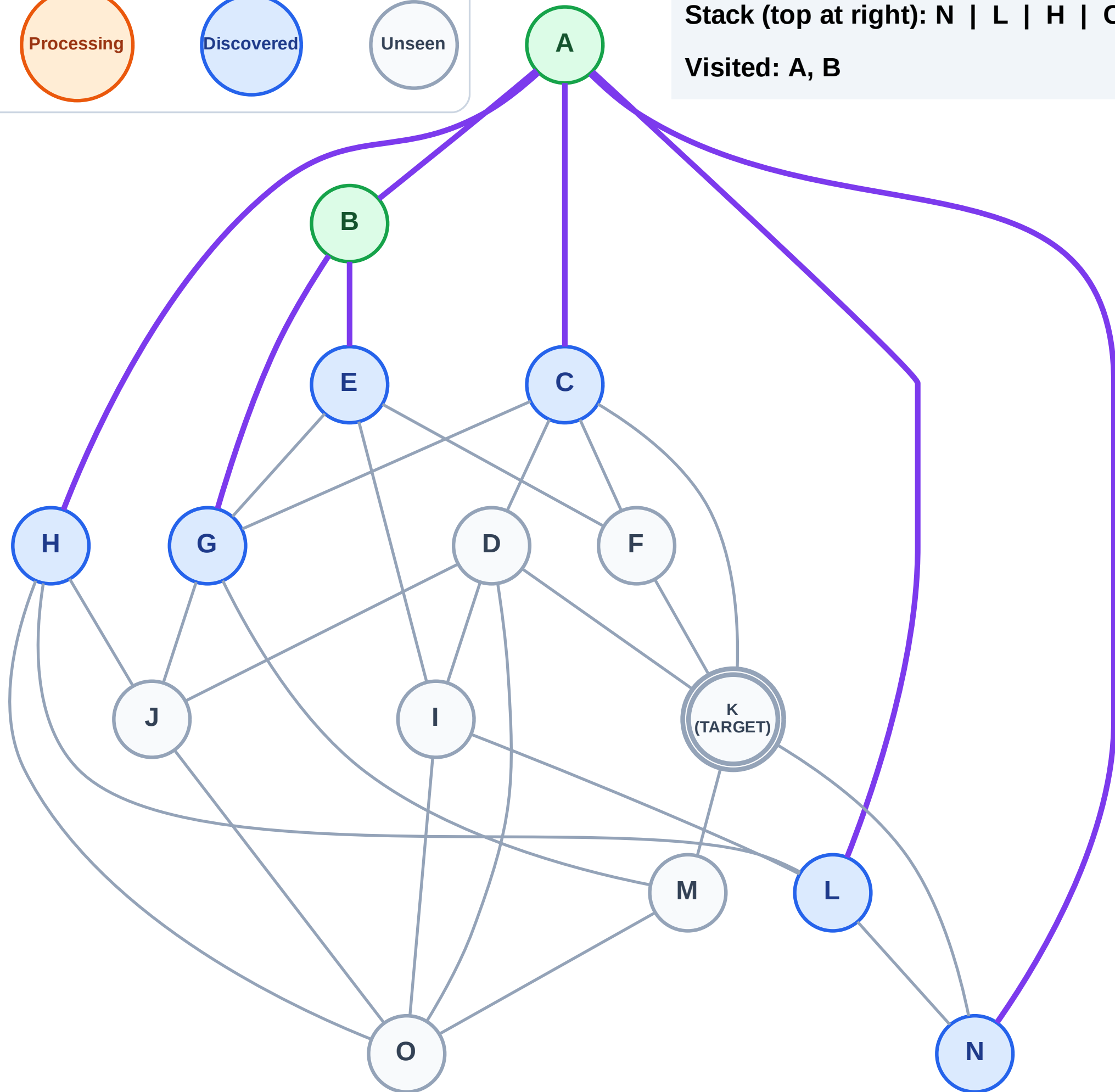
Step 5: Discover G, E from B

Legend



Stack (top at right): N | L | H | C | G | E

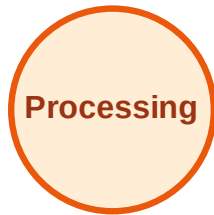
Visited: A, B



DFS Traversal

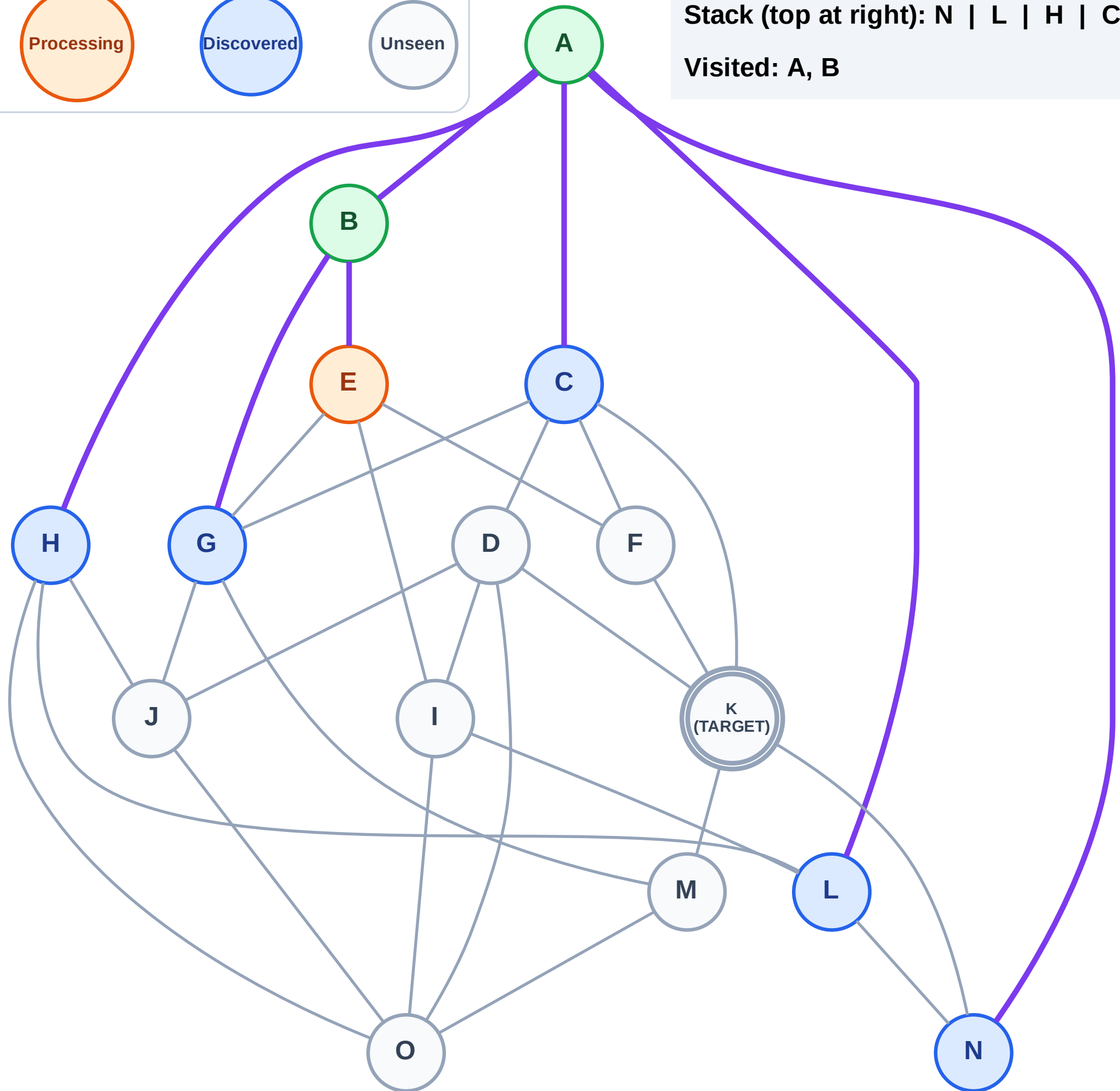
Step 6: Process node E

Legend



Stack (top at right): N | L | H | C | G

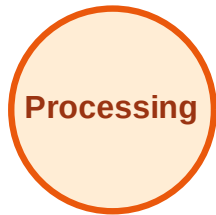
Visited: A, B



DFS Traversal

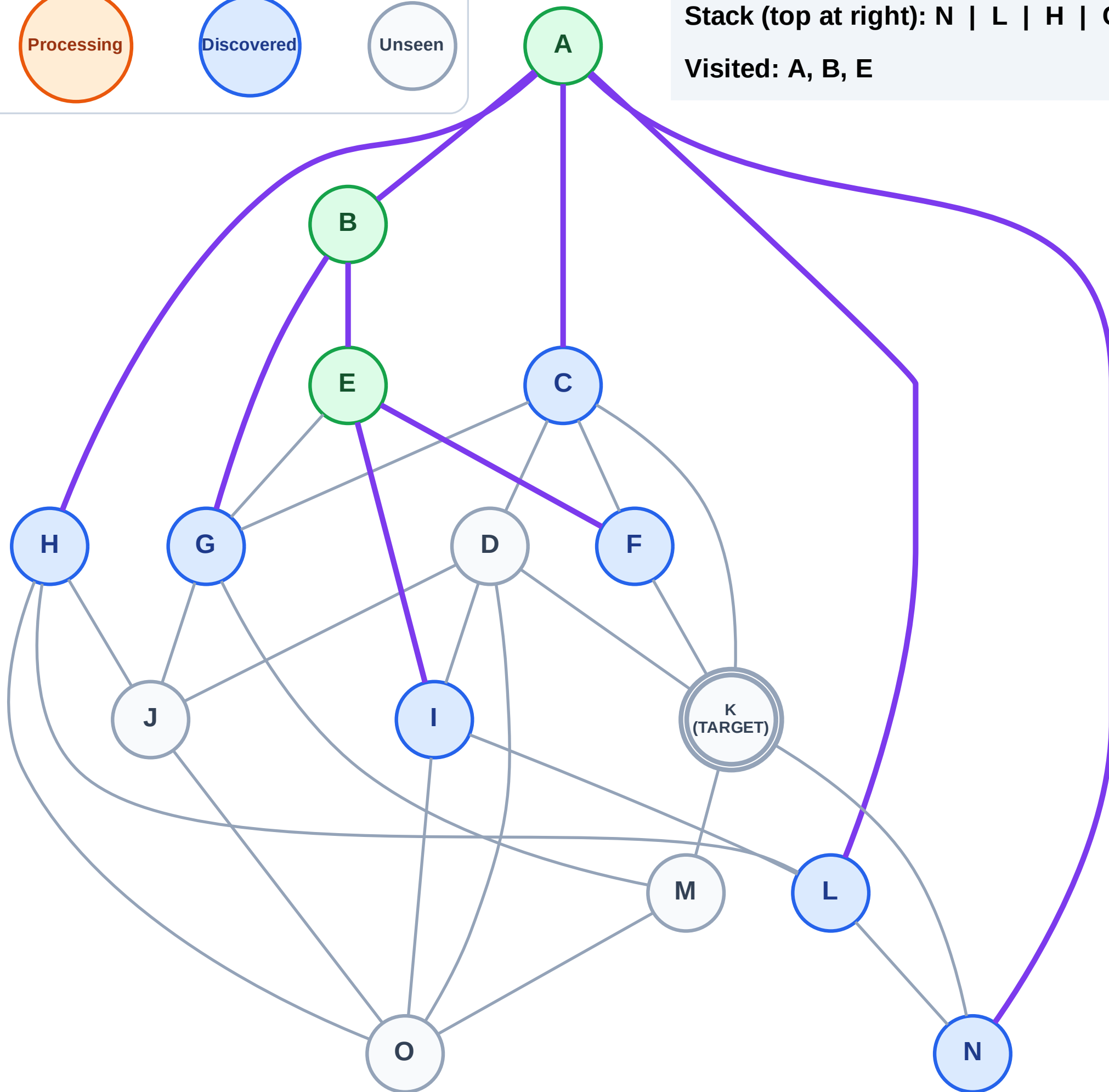
Step 7: Discover I, F from E

Legend



Stack (top at right): N | L | H | C | G | I | F

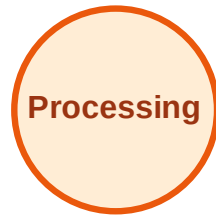
Visited: A, B, E



DFS Traversal

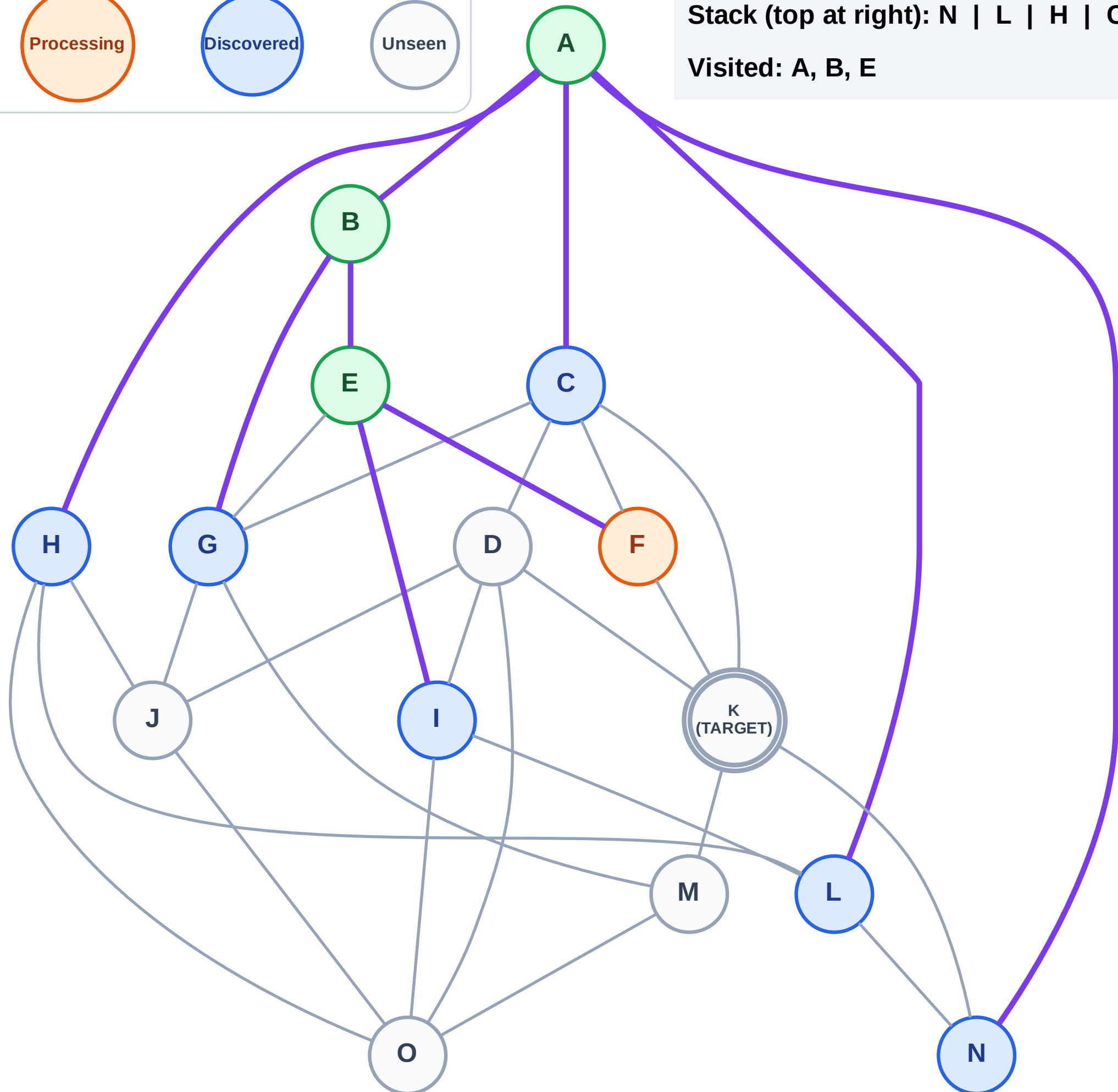
Step 8: Process node F

Legend



Stack (top at right): N | L | H | C | G | I

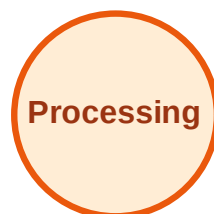
Visited: A, B, E



DFS Traversal

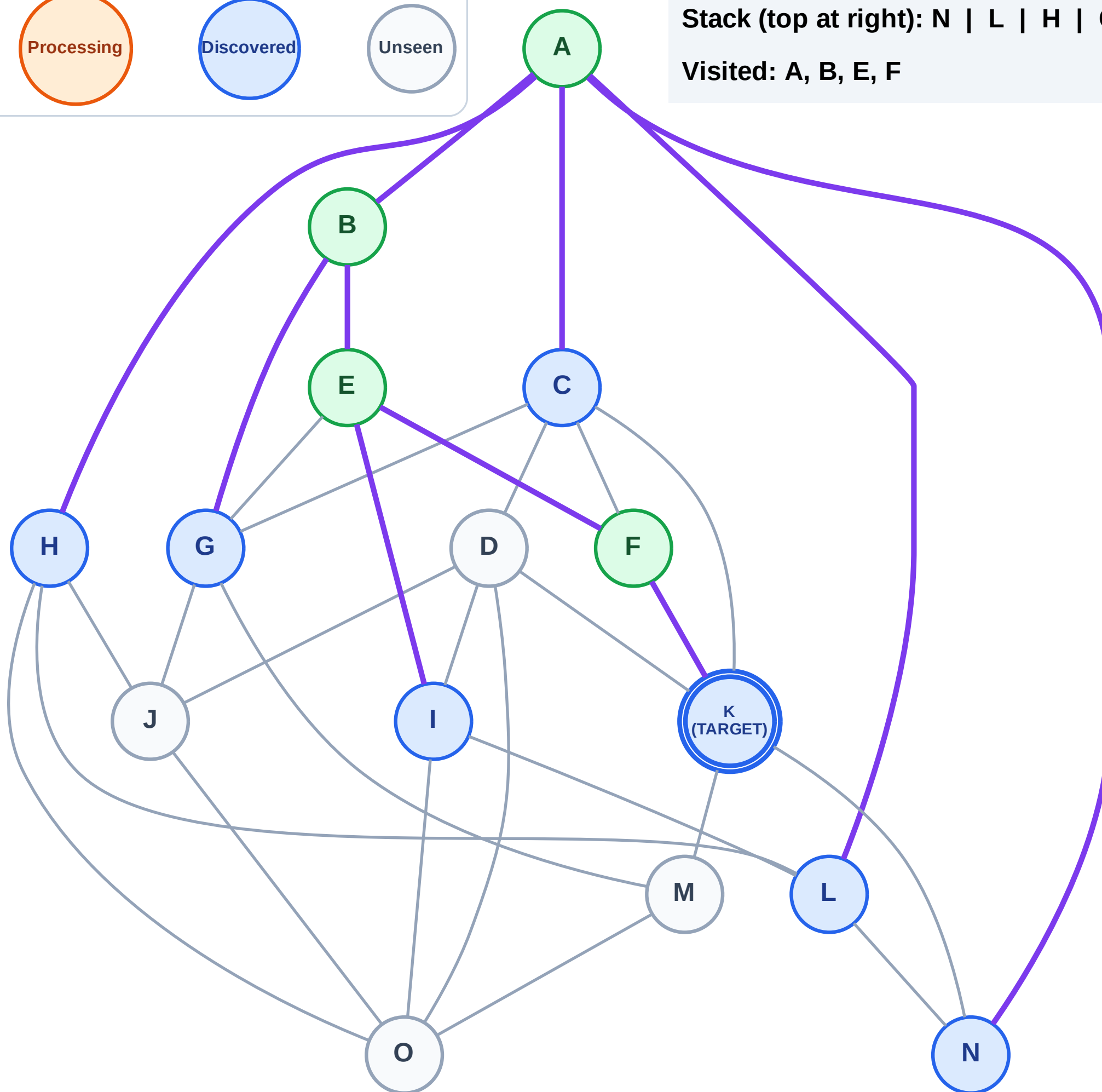
Step 9: Discover K from F

Legend



Stack (top at right): N | L | H | C | G | I | K

Visited: A, B, E, F



DFS Traversal

Step 10: Process node K

Legend

Complete

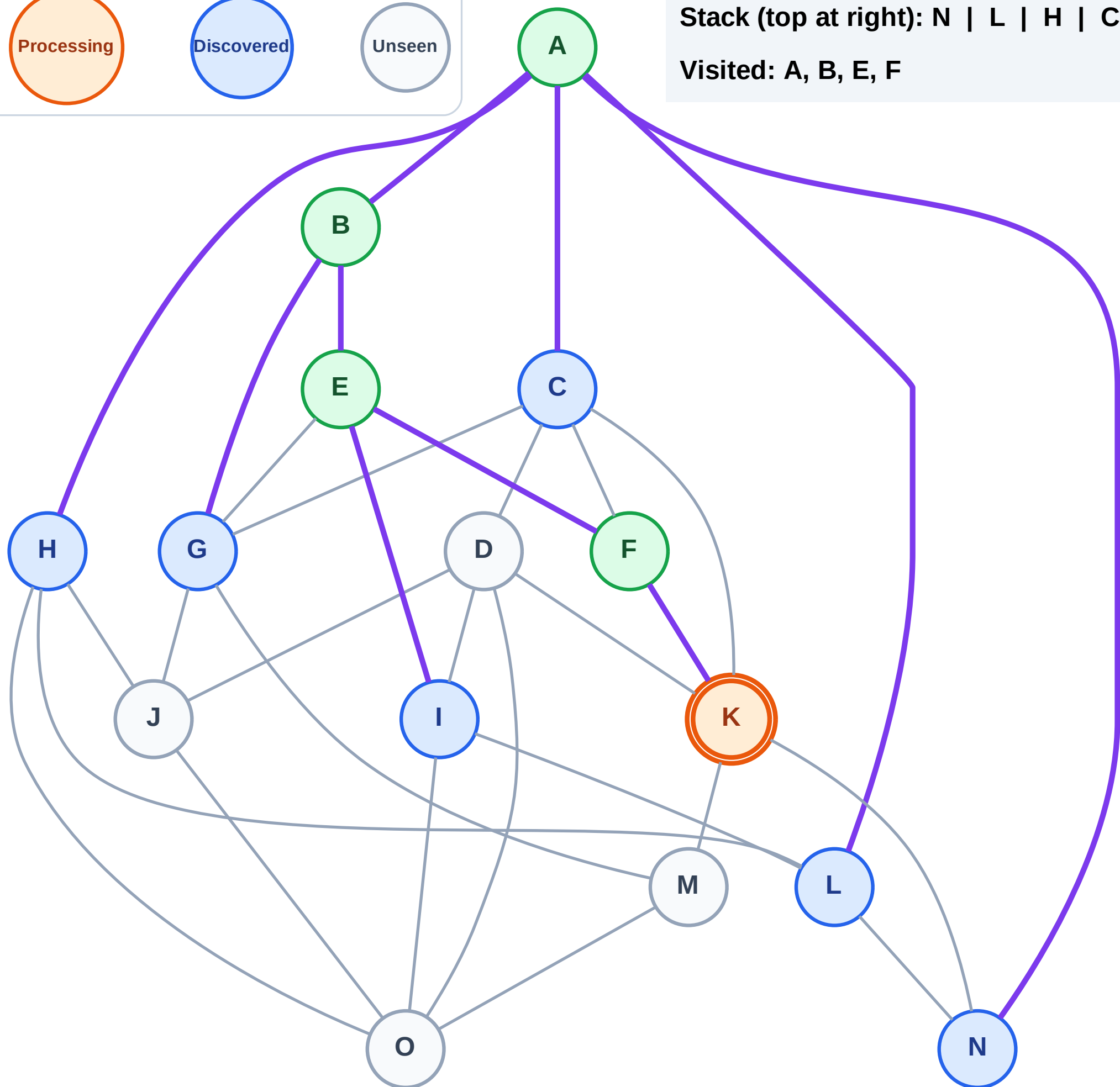
Processing

Discovered

Unseen

Stack (top at right): N | L | H | C | G | I

Visited: A, B, E, F



DFS Traversal

Step 11: TARGET NODE K REACHED! Path (4 hops): A → B → E → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): N | L | H | C | G | I

Visited: A, B, E, F, K

